

Integrated Circuit ECE326

Lec. 13

Layout

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- ☐ **Living in a 3D World**
- ☐ **Design Rules Check**
- ☐ **Geometric Design Rules**
- ☐ **Layout Methodologies**
- ☐ **Stick diagram**
- ☐ **Exersice_01**
- ☐ **Exersice_02**
- ☐ **Exersice_03**
- ☐ **Exersice_04**



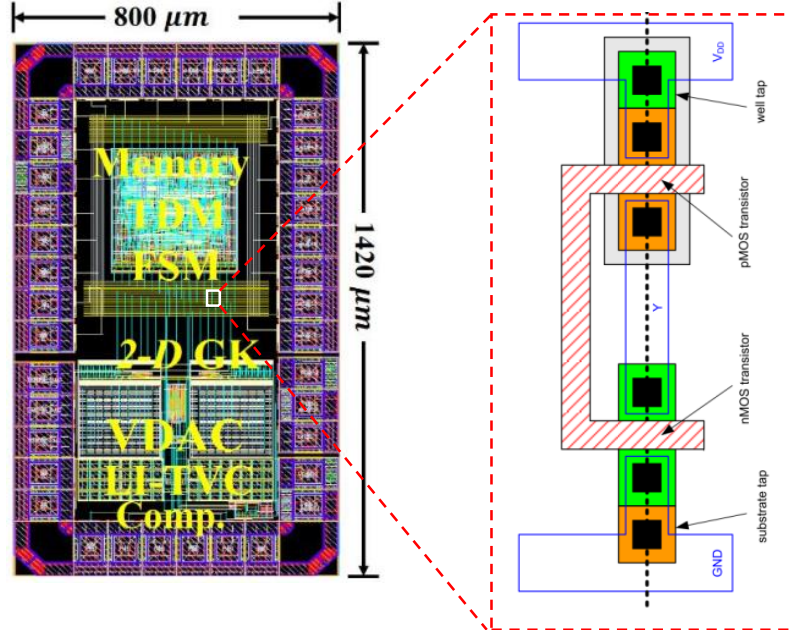
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- ☐ **Exercise_04**

Living in a 3D World (1/2)



□ Your Job

□ Create Layout

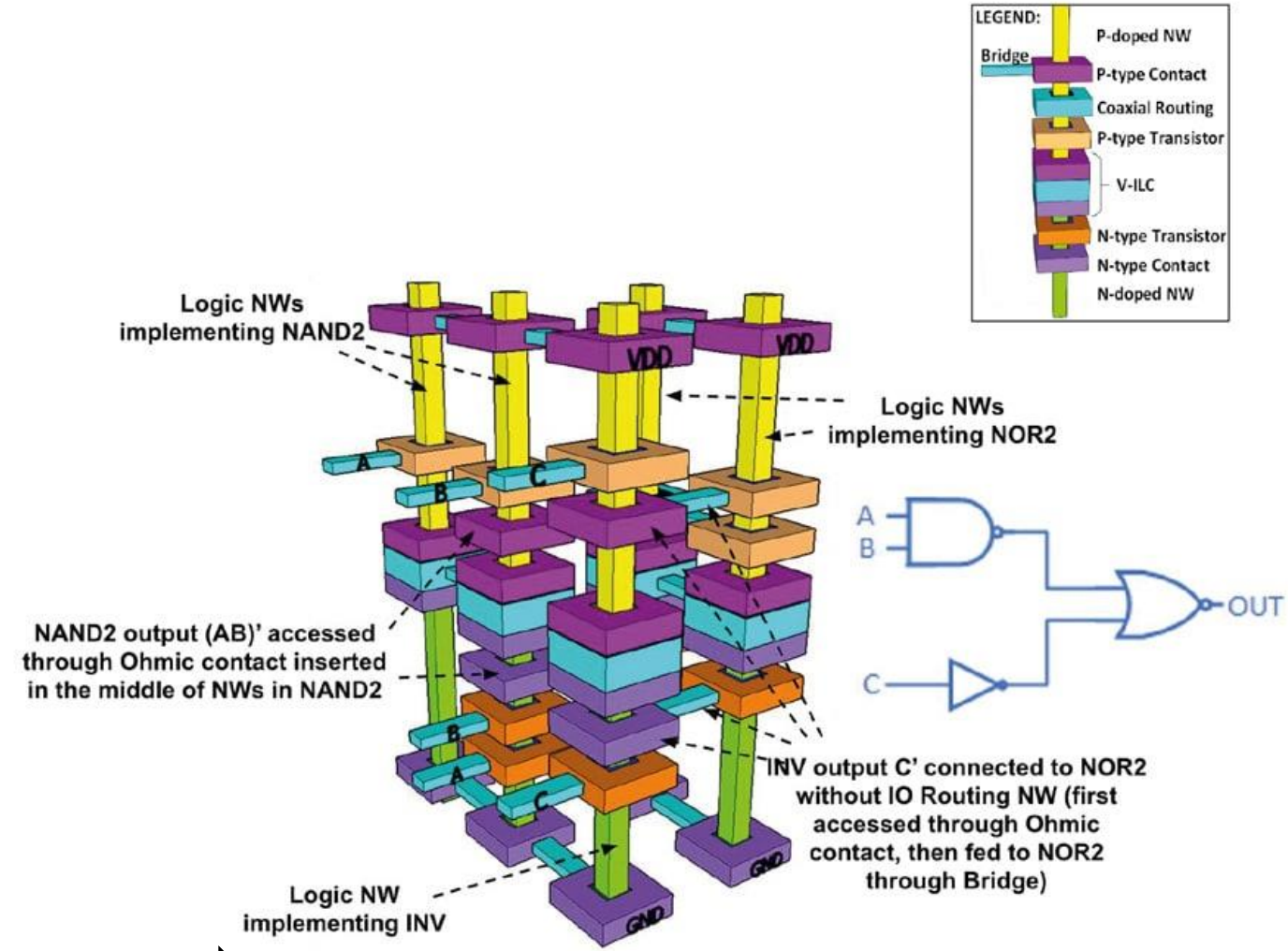


□ Manufacturer's task

□ Fabricate the Layout



Fabrication process controls
vertical dimensions



*Circuit for a combinational
logic function with NAND, NOR
and INV logic gates.*

Living in a 3D World (2/2)



- ❑ Layout is a group of planar geometries drawn on different **colored** layers corresponding to different masks used in fabrication (**Each layer has unique color**)
- ❑ **Planar dimensions** are mainly set by design through mask dimensions.
- ❑ Designer is typically concerned with both **schematic** and **layout** “views” of the design .
- ❑ Once layout is completely **done**, the tool generate a special GDSII format (**tapeout**) to send to IC fabrication facility.

Layer	Color	Representation
Well (p,n)	Yellow	
Active Area (n+,p+)	Green	
Select (p+,n+)	Green	
Polysilicon	Red	
Metal1	Blue	
Metal2	Magenta	
Contact To Poly	Black	
Contact To Diffusion	Black	
Via	Black	

Depending on fab & process

Fabrication process also sets some constraints on planar dimensions
(**Design Rules Check**)

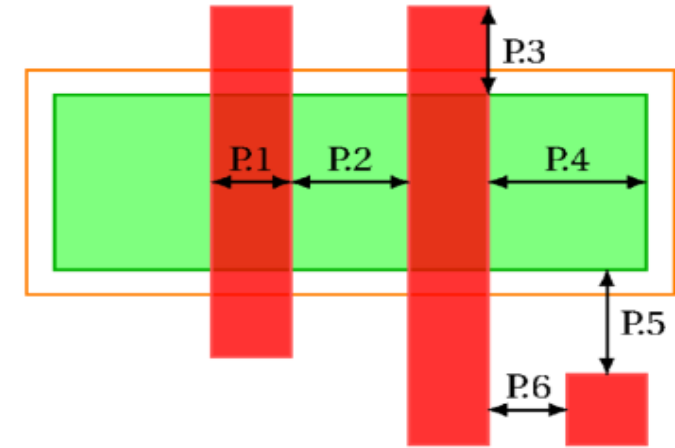


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Design Rules Check (1/5)



- ❑ Design Rules Check (**DRCs**) are the interface between the designer and process engineer.
- ❑ They provide guidelines for constructing process masks .
- ❑ These are generally **categorized** as:
 - ❑ **Intra-layer rules**: minimum widths, spacing, area, etc., only relating to a single layer.
 - ❑ **Inter-layer rules**: minimum enclosures, extensions, overlaps,... etc., between two layers.
 - ❑ **Special rules**: non geometric rules, such as antenna rules, density, distance to well tap, ...etc.



Poly	Rule name (minimum)	Length
P.1	Poly width	65nm
P.2	Space poly and active	80nm
P.3	Poly extension beyond active	30nm
P.4	Enclosure active around gate	85nm
P.5	Space field poly to active	70nm
P.6	Space field poly	25nm

We can create wider or longer transistors, ploy, metal But you waste silicon area and generate parasitic elements

Design Rules Check (2/5)



XH018 BASIC DESIGN RULES

Mask	width [μm]	Spacing [μm]
N-well	0.86	1.4
Active Area	0.22	0.28
Poly-silicon Gate	0.18	0.25
Poly-silicon Resistor	0.44	0.44
Contact	0.22	0.25
Metal 1	0.23	0.23
Via 1, 2, 3, 4	0.26	0.26
Metal 2, 3, 4, 5	0.28	0.28
Top Via	0.36	0.35
Top Metal	0.44	0.46
Thick Metal	3.0	2.5

XH035 BASIC DESIGN RULES

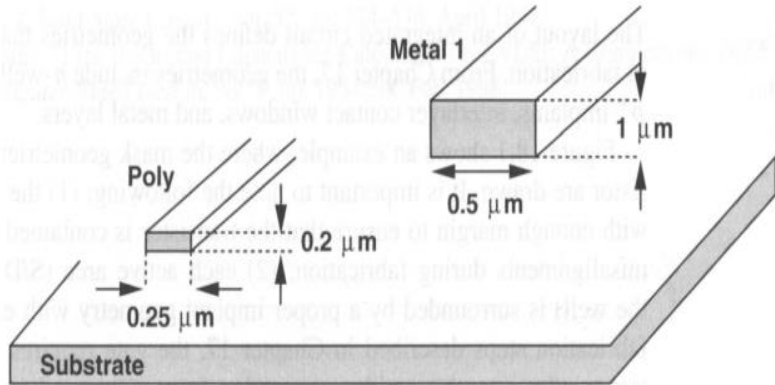
Mask	width [μm]	Spacing [μm]
N-well	1.6	1.0
Active Area	0.5	0.6
Poly-silicon Gate/Resistor	0.35	0.45
Contact	0.4	0.4
Metal 1, Via 1, 2, 3	0.5	0.45
Metal 2, 3	0.6	0.5 (0.6 if Top Layer Metal)
Metal 4	0.6	0.6
Thick Metal	3.0	2.5



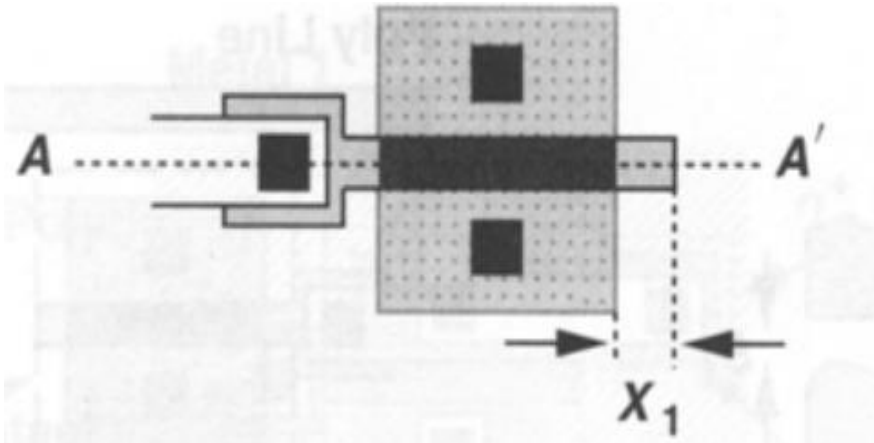
0.18 μm Process Family:
> XH018

0.35 μm Process Family:
> XH035

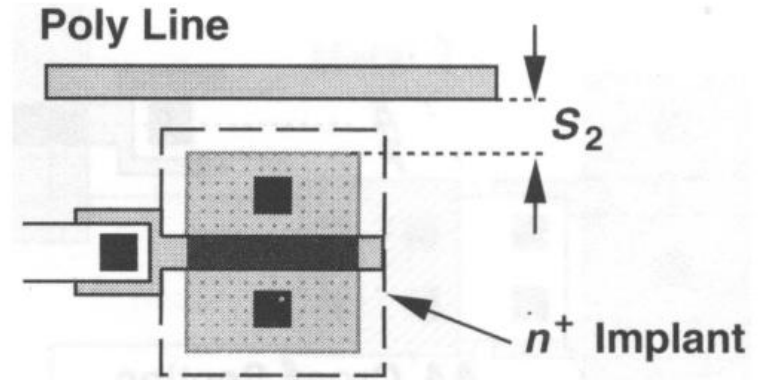
Design Rules Check (3/5)



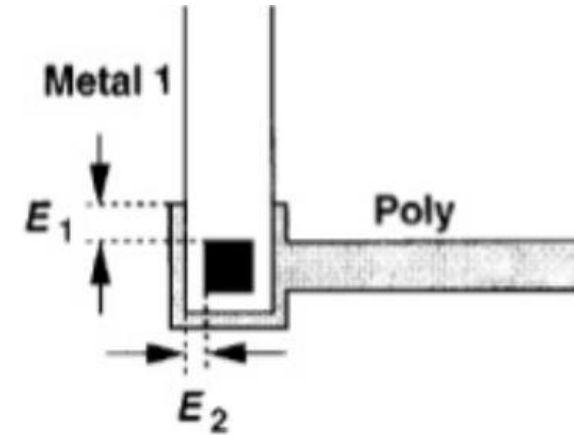
Minimum Width Rules



Minimum Extension Rules



Minimum Spacing Rules

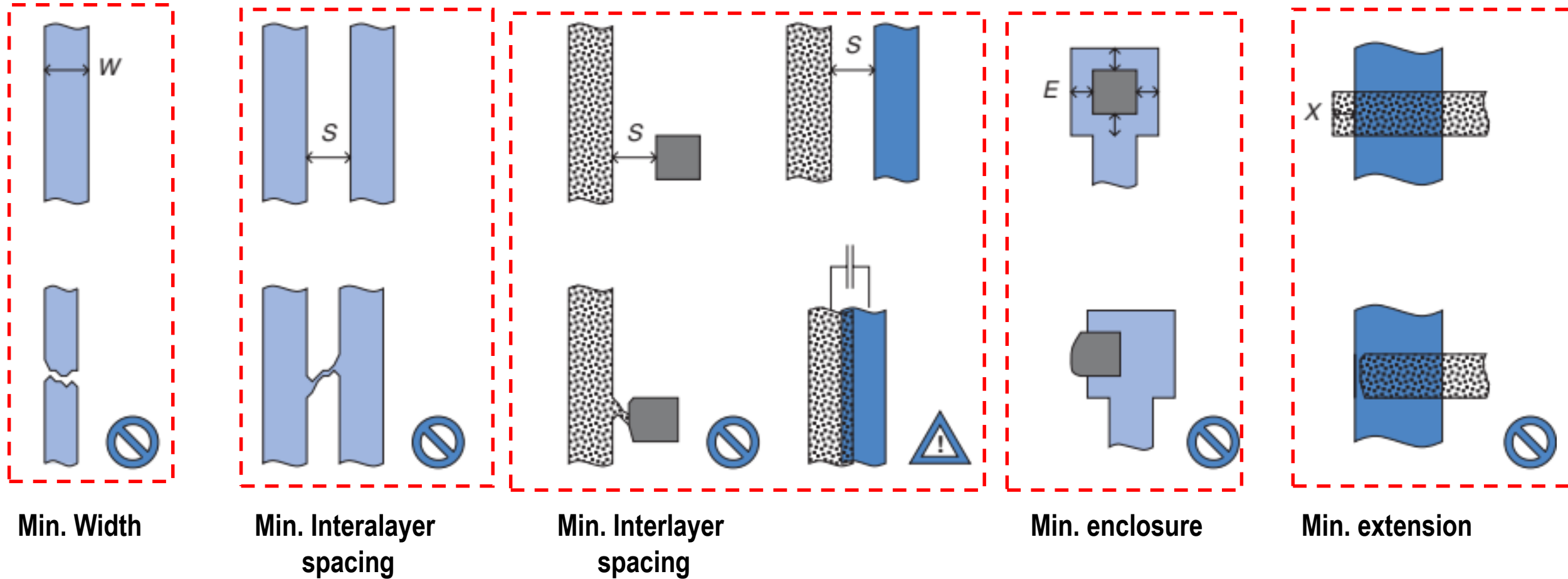


Minimum Enclosure Rules

Design Rules Check (4/5)



❑ What happens to transistor if min extension of poly is not observed.



Design Rules Check (5/5)



- ☐ In practical, DRCs of the layout are **very complicated**.
- ☐ Checked by a software CAD tool (industry standard DRC tool)
 - ☐ **Assura** (Cadenc) 
 - ☐ **Calibre** (Mentor Graphics) 
 - ☐ **Hercules** (Synopsys) 
- ☐ Generally, the **foundry** will not accept any design with DRC **errors**
 - ☐ All errors must be fixed
 - ☐ In special cases, minor DRC errors can be accepted by the foundry (but you must request an official DRC **error waiver**)



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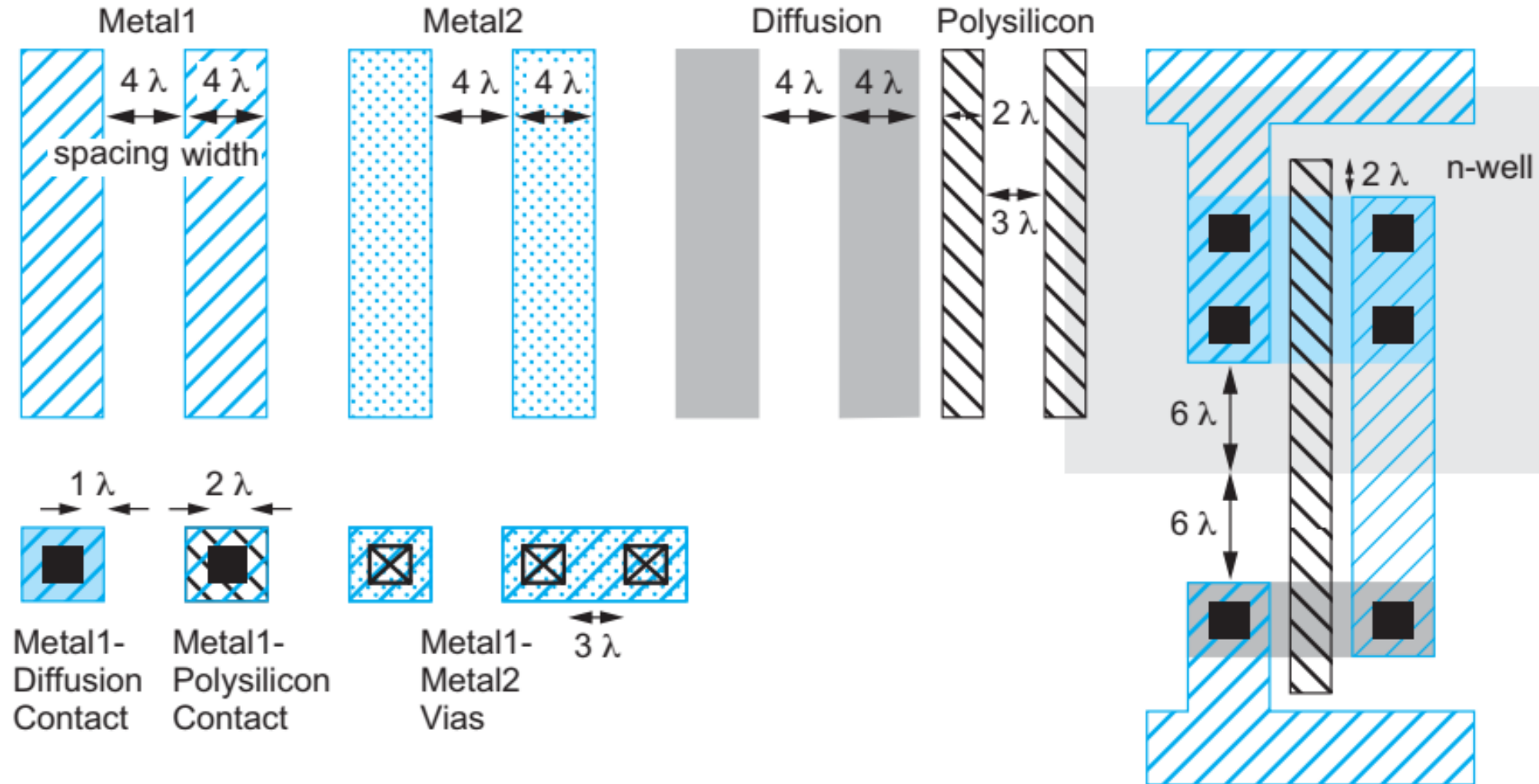
Geometric Design Rules (1/2)



- ❑ Two ways to write the geometric design rules:

Absolute rules	Scalable rules
❑ Design rules can be expressed in absolute physical units. Ex. poly width $0.3\mu\text{m}$, poly spacing $0.45\mu\text{m}$, metal width $0.45\mu\text{m}$ ❑ Typically not multiples of one another in order to get the densest layout ❑ Difficult to remember, design takes more time and not scalable ❑ Difficult to port from one process to another ❑ layout should be repeated from scratch for newer technology	❑ Scalable design rules express dimension in normalized units called lambda (λ) ❑ Normalize everything so the minimum gate length (i.e. width of poly gate) is $2*\lambda$ =feature size(f) ❑ All other design rules are expressed in integer multiples of λ. Ex. poly width 2λ , space 3λ, metal width & space 3λ ❑ These are subsequently scaled to generate masks for a variety of processes

Geometric Design Rules (2/2)



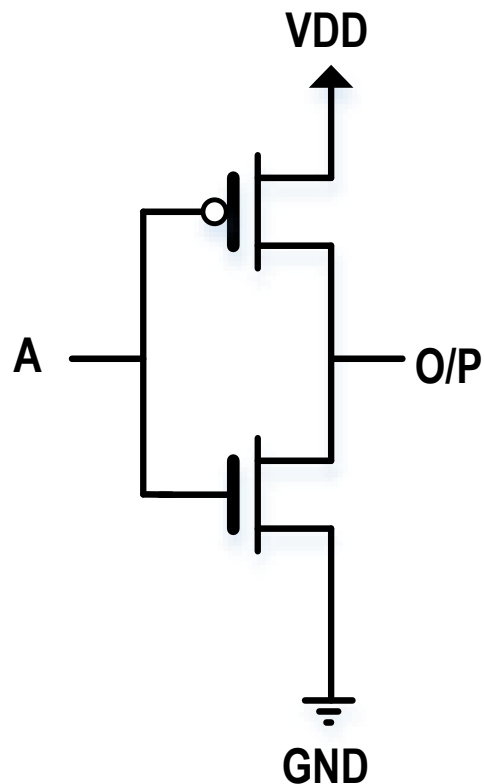


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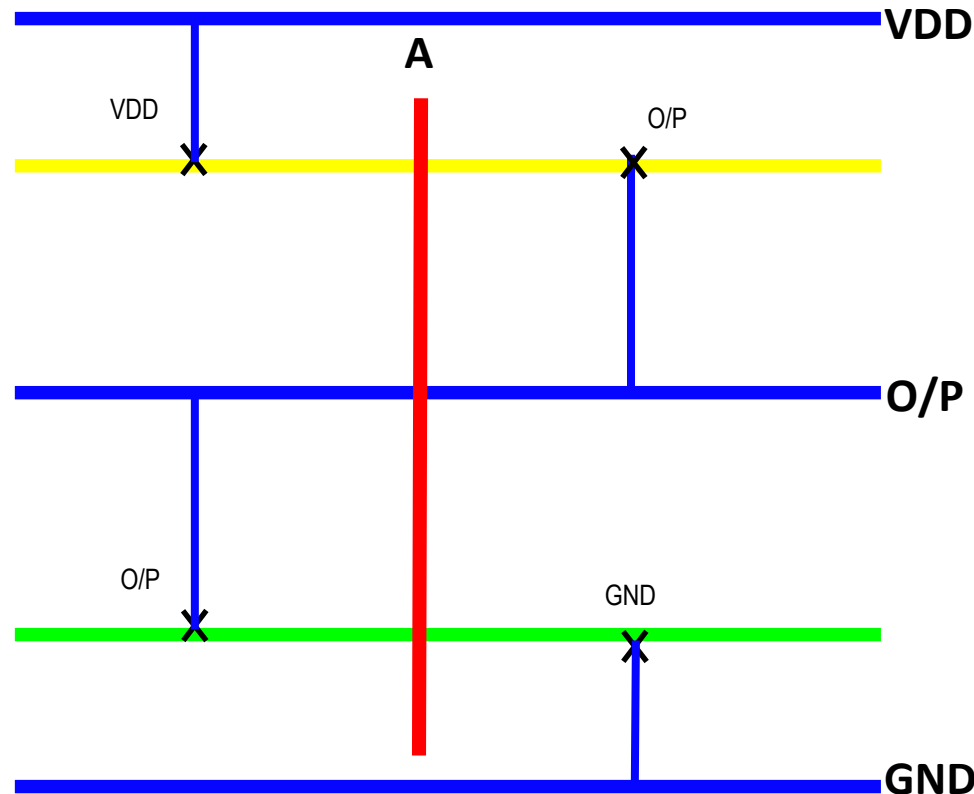
Layout Methodologies



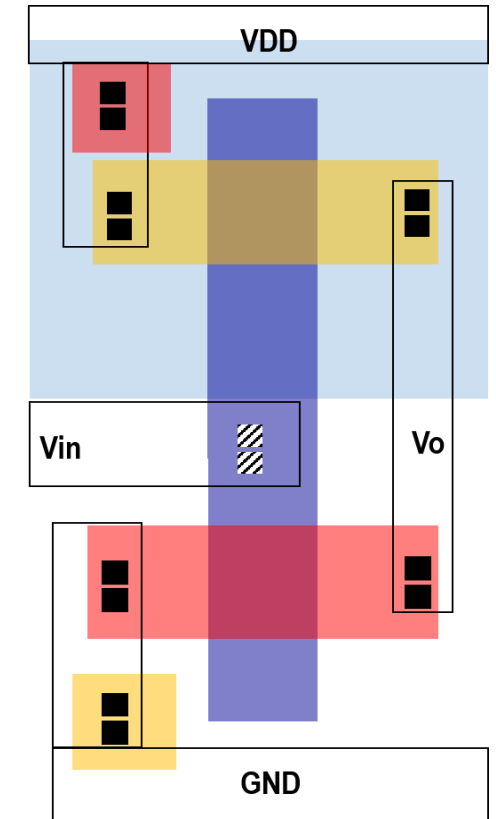
- ❑ Layout can be very time consuming to fit gates/elements together nicely
 - ❑ Stick diagram
 - ❑ Standard cell design methodology



Schematic Circuit



Stick Diagram



Standard Cell design



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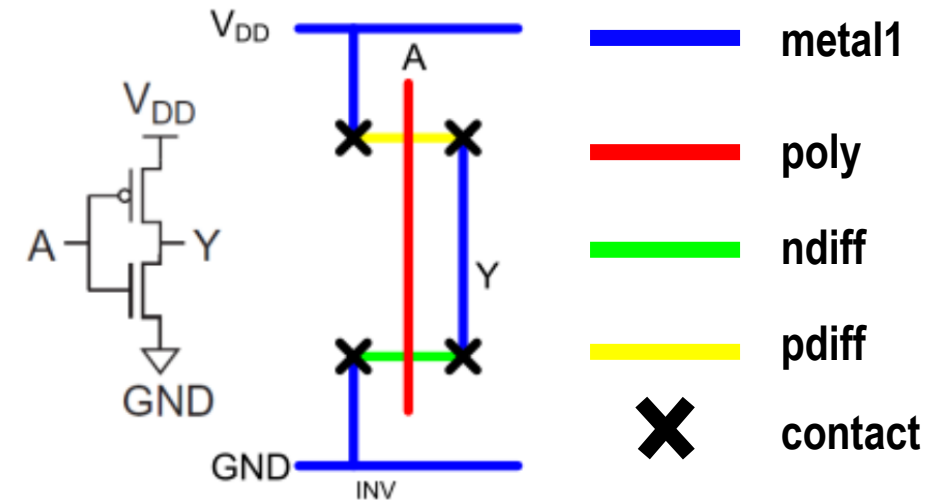
Stick diagram (1/2)



- ❑ A stick diagram is a symbolic layout to plan layout **quickly**.
- ❑ Wires are drawn as stick figures with **no width, so no need to scale**
- ❑ It is always much **faster** to design layout on paper using stick diagram **first** before using the layout CAD tool

❑ Important Tips (Do)

- ❑ Metal1 **GND** rail at **bottom**
- ❑ Metal1 **V_{DD}** rail at **top**
- ❑ Metal1 makes **connections**
- ❑ **Horizontal** N-diffusion and p-diffusion strips
- ❑ **Vertical** polysilicon gates
- ❑ Transistor = intersection of poly and diffusion
- ❑ nMOS usually are placed near **GND**, and pMOS near **V_{DD}**
- ❑ **Must** connect ndiff to metal and then metal to pdiff (No direct connection)



Does not show more information about masks of the layout

Stick diagram (2/2)



- ❑ Step1:- Construct the logic diagram of the given function (pull-up and pull-down networks)
- ❑ Step2:- Identify each **Node** between transistors and give them a unique name (Main Nodes: V_{DD} , O/P, GND and Sub-Nodes: 1,2,3...)
- ❑ Step3:-Draw the **Euler diagram**.
- ❑ Step4:-Identify the **Euler path** for pull-up and pull-down networks
 - ❑ **Nodes can be repeated** but **Paths cannot be repeated**.
 - ❑ The path should be **valid** in both Pull-up and pull-down networks.
- ❑ Step5:-Draw the stick diagram
 - ❑ Align V_{DD} , O/P, GND (**Horizontal lines**)
 - ❑ Align pdiff and ndiff (**Horizontal lines**)
 - ❑ Align Euler path (**Vertical lines**)
 - ❑ Identify the main nodes and sub-nodes (X)
 - ❑ **Connect** the main Nodes and then Sub nodes.

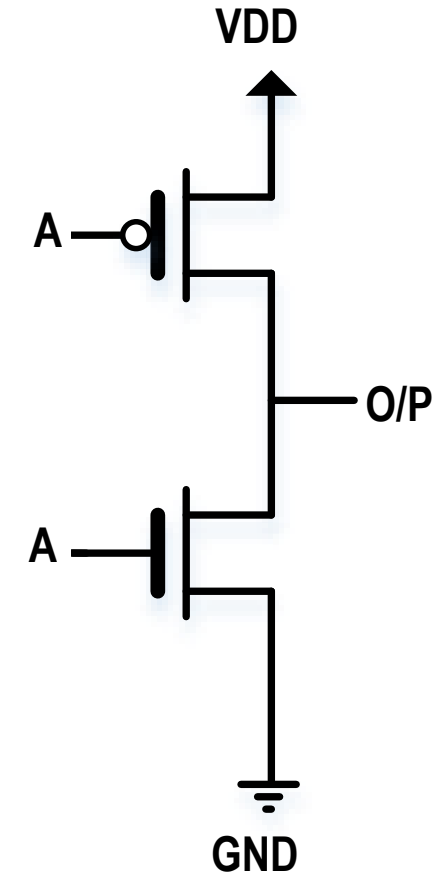


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Exercise_01



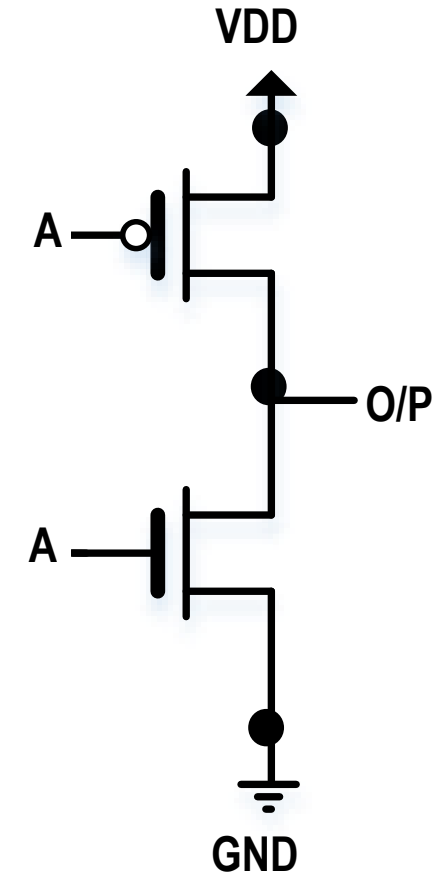
- Draw the stick diagram of the following circuit (CMOS inverter)
- Steps1:- Construct the logic diagram of the given function (pull-up and pull-down networks) → **given**



Exercise_01



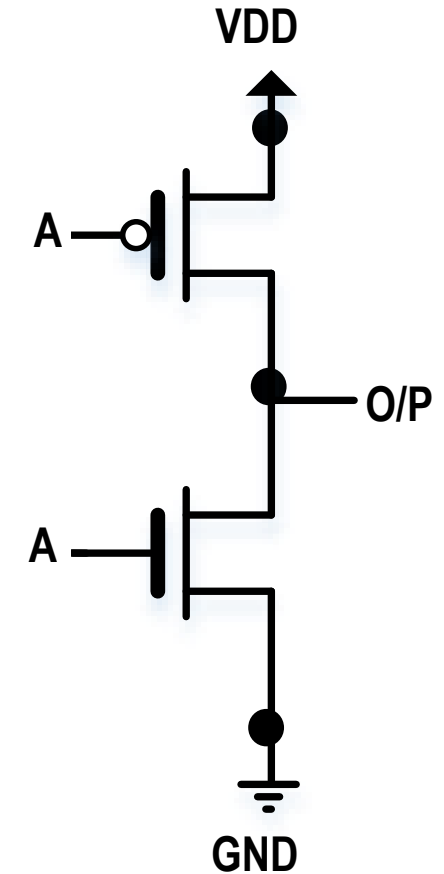
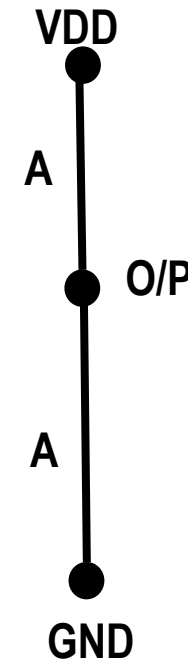
- ☐ Draw the stick diagram of the following circuit (CMOS inverter)
- ☐ Step2:- Identify each connection between transistors only and give them a unique name.
 - ☐ Main Nodes: VDD, O/P, GND
 - ☐ Sub-Nodes: **there is no sub node**



Exercise_01



- ☐ Draw the stick diagram of the following circuit (CMOS inverter)
- ☐ Step3:- Draw the Euler diagram



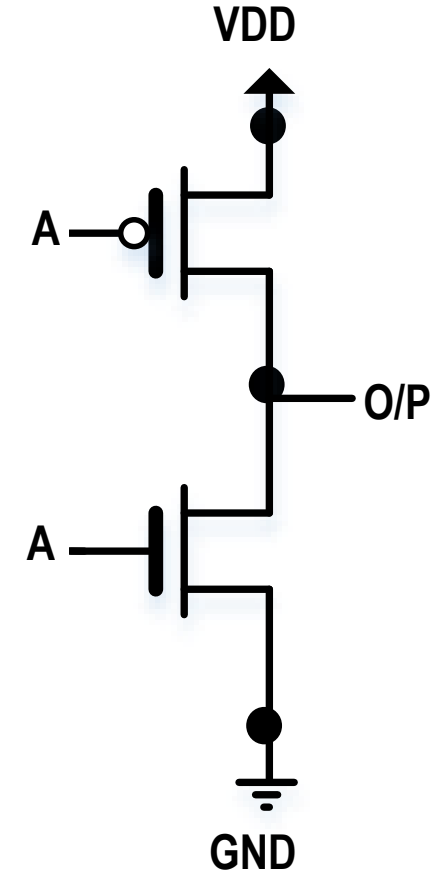
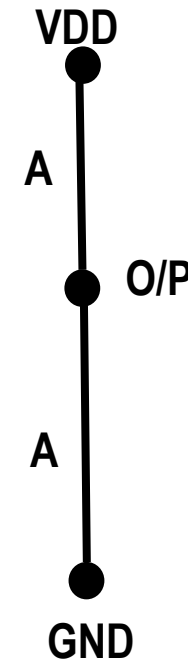
Exercise_01



- ☐ Draw the stick diagram of the following circuit (CMOS inverter)
- ☐ Step4:- Identify the Euler path

Condition of Euler Path

1. Nodes can be repeated but Paths cannot be repeated.
2. The path should be valid in both Pull-up and pull-down networks.



Exercise_01

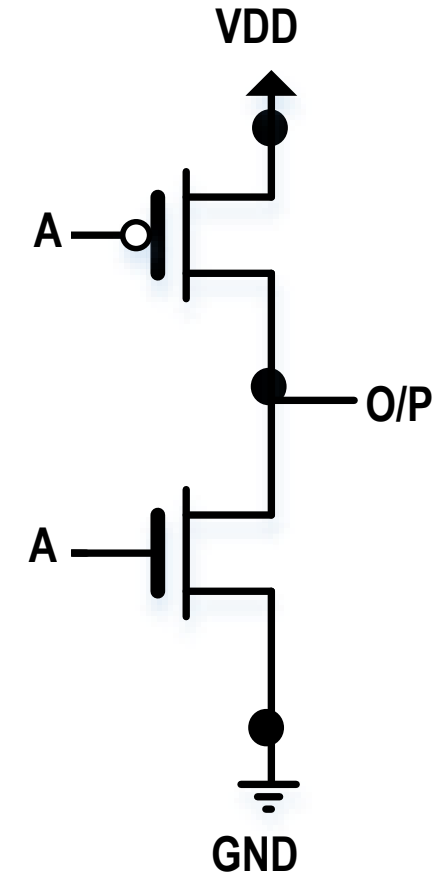
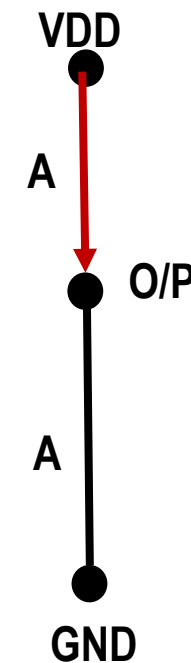


☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step4:- Identify the Euler path

Euler path for **Pull-up** network is

A



Exercise_01



☐ Draw the stick diagram of the following circuit (CMOS inverter)

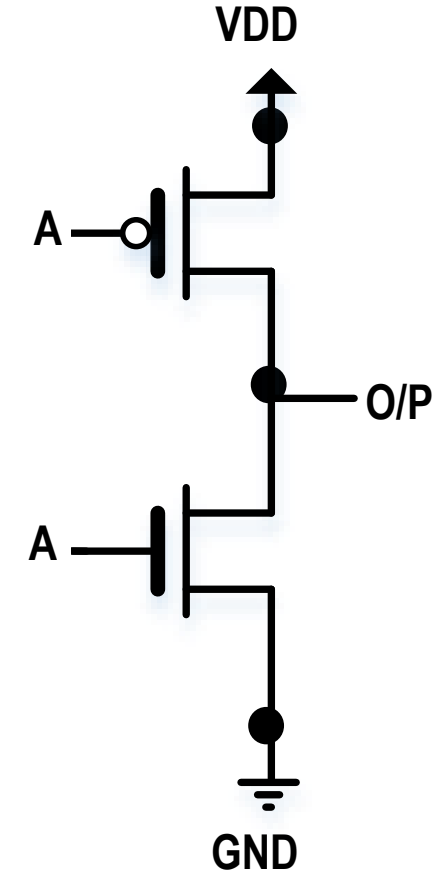
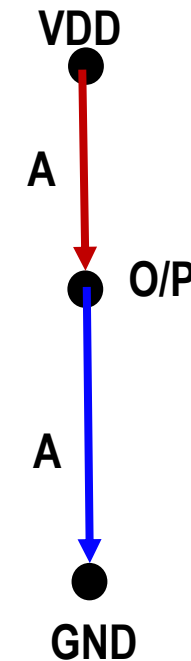
☐ Step4:- Identify the Euler path

Euler path for Pull-up network is

A

Euler path for Pull-down network is

A



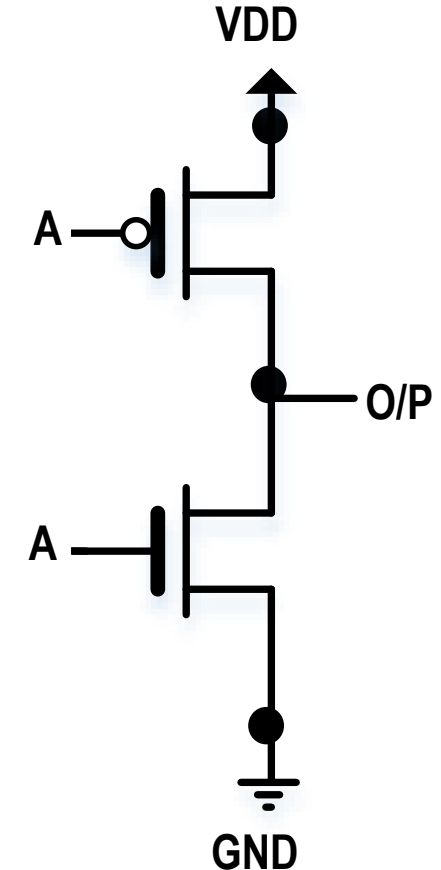
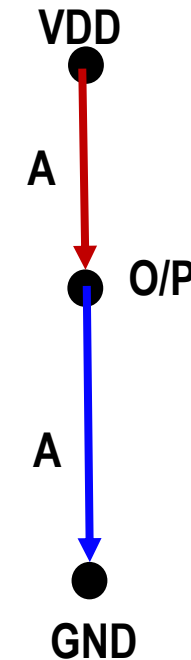
Exercise_01



☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step5:- Draw the stick diagram

A (in order)



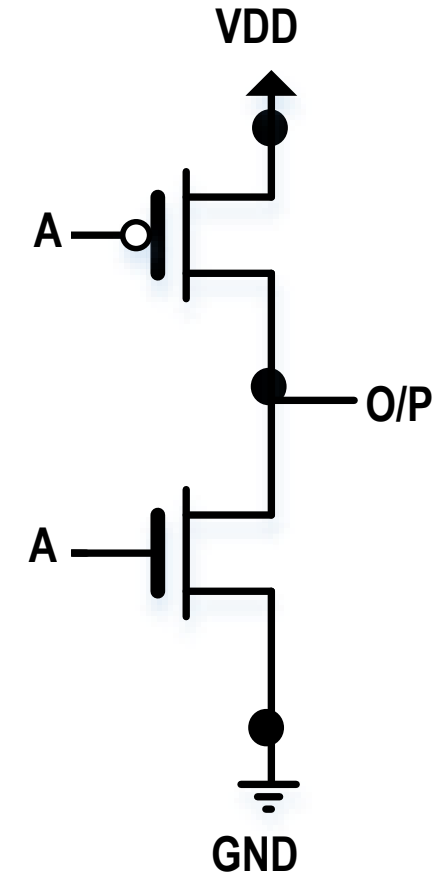
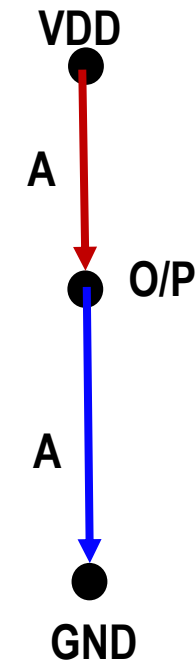
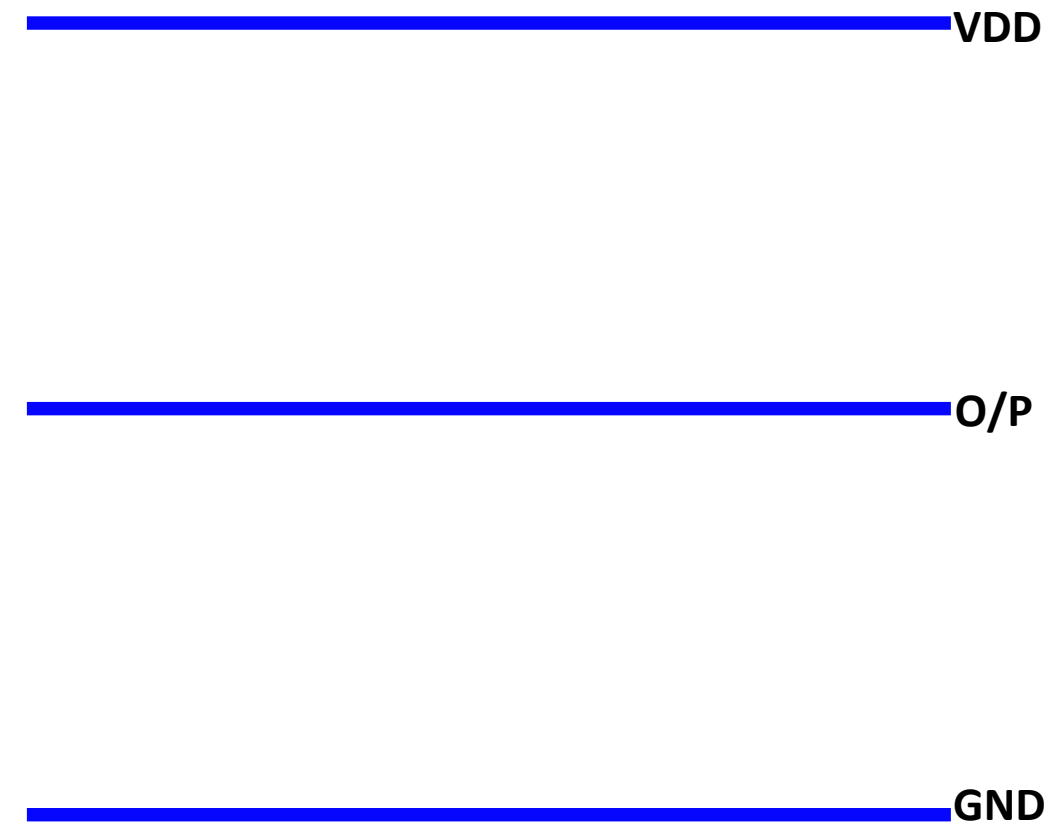
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☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step5:- Draw the stick diagram

A (in order)



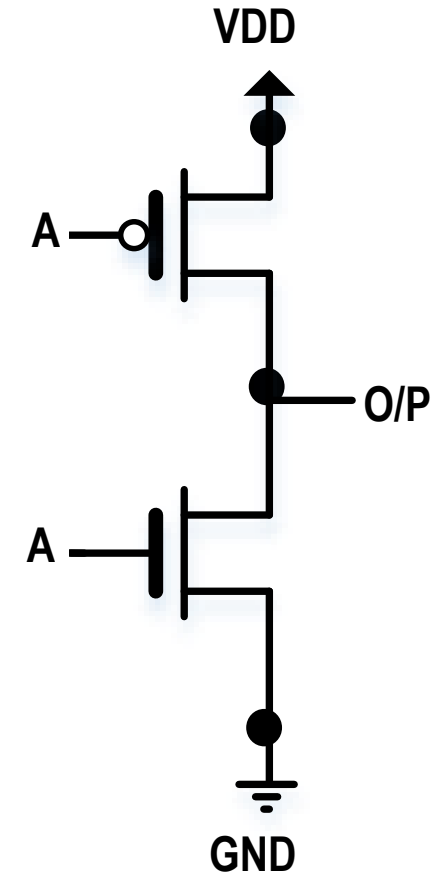
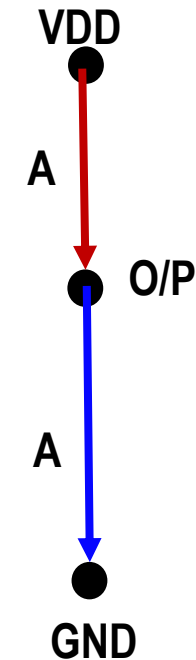
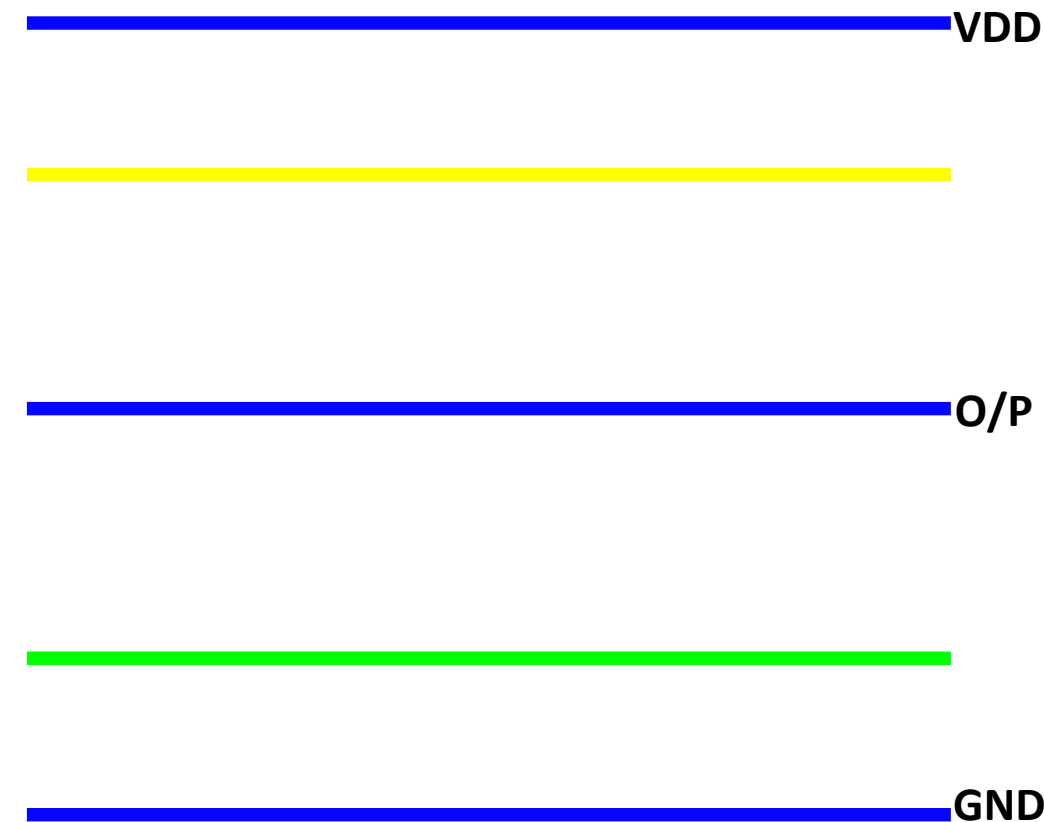
Exercise_01



☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step5:- Draw the stick diagram

A (in order)



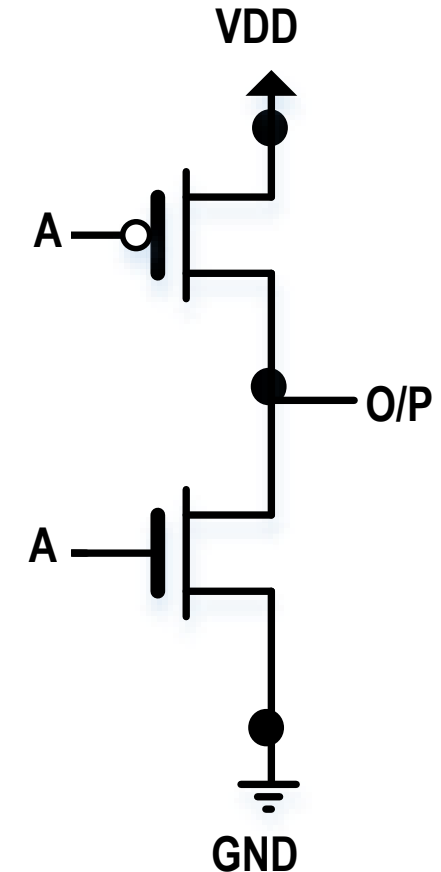
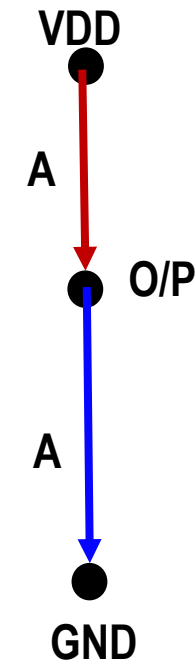
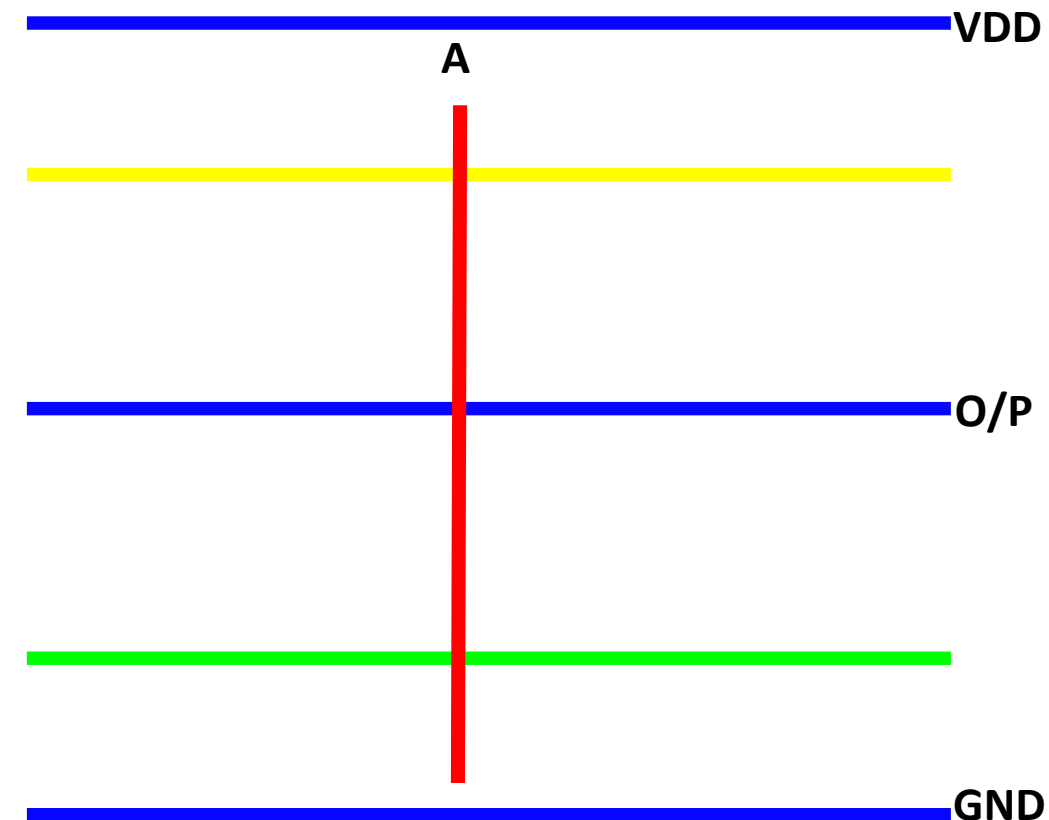
Exercise_01



☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step5:- Draw the stick diagram

A (in order)



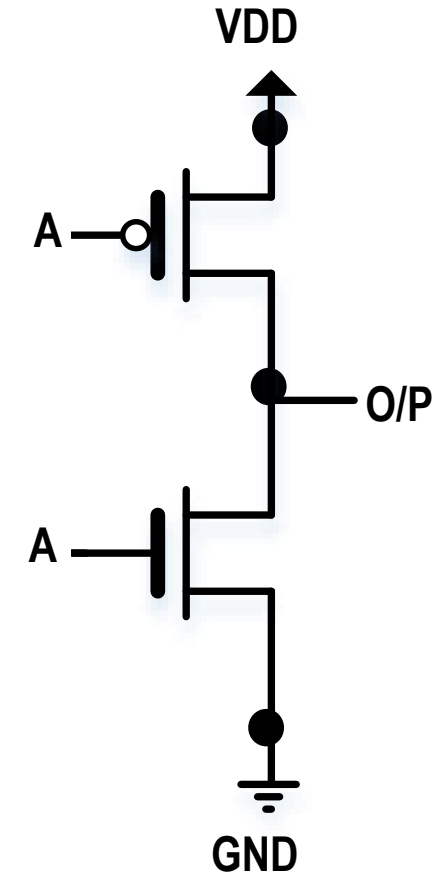
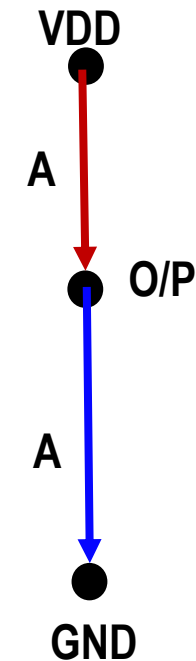
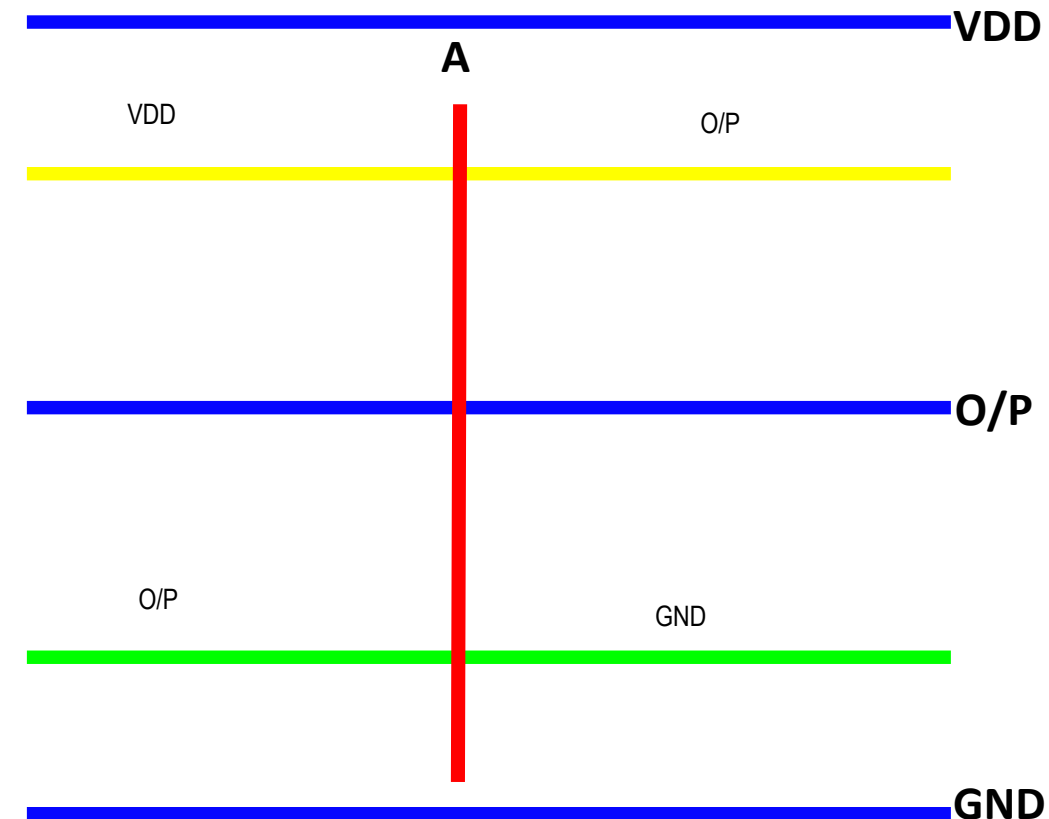
Exercise_01



☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step5:- Draw the stick diagram

A (in order)



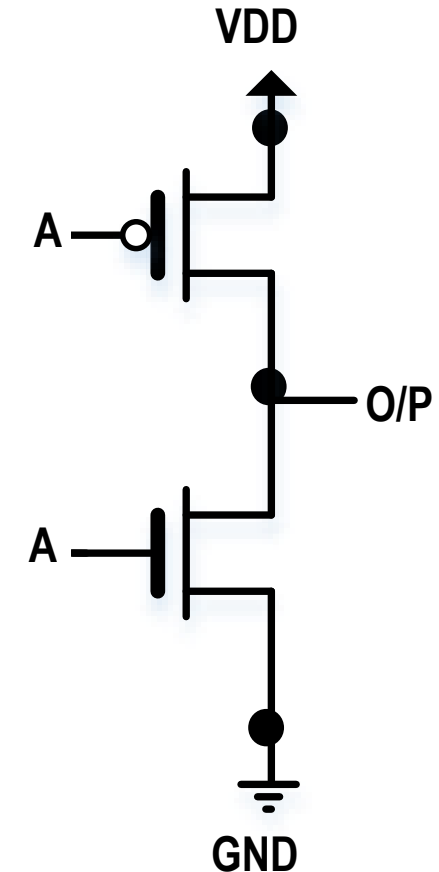
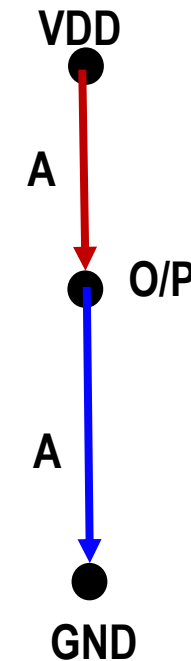
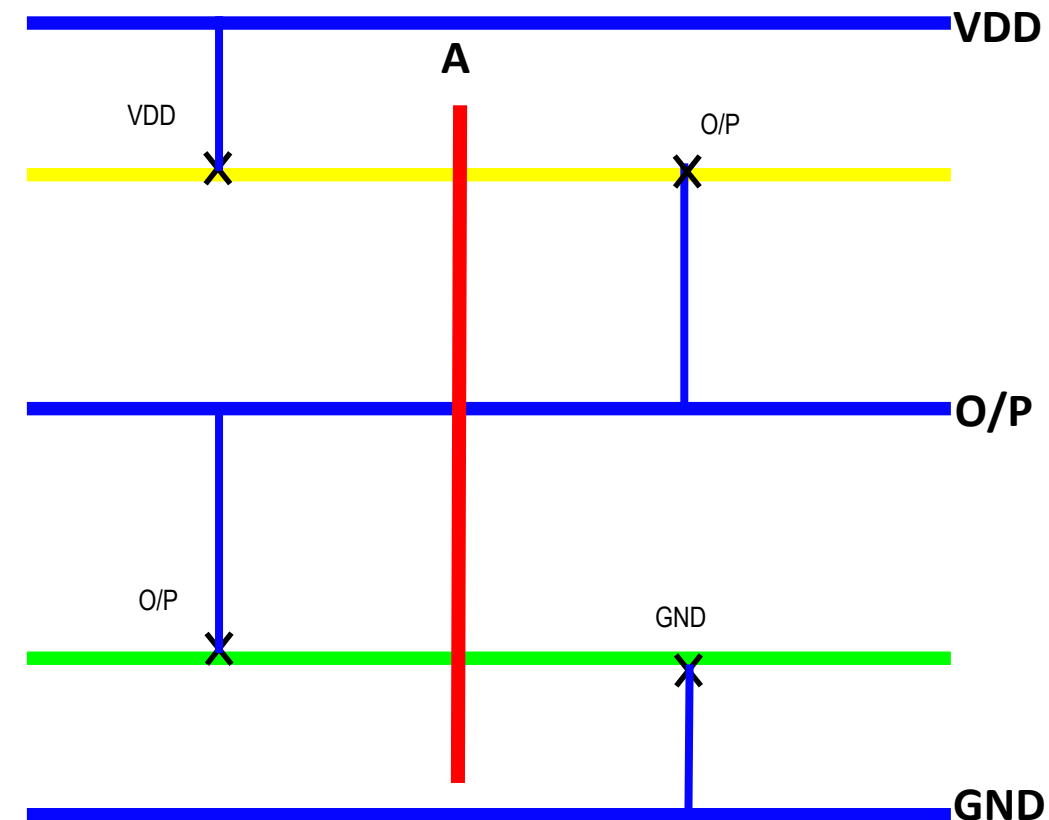
Exercise_01



☐ Draw the stick diagram of the following circuit (CMOS inverter)

☐ Step5:- Draw the stick diagram

A (in order)



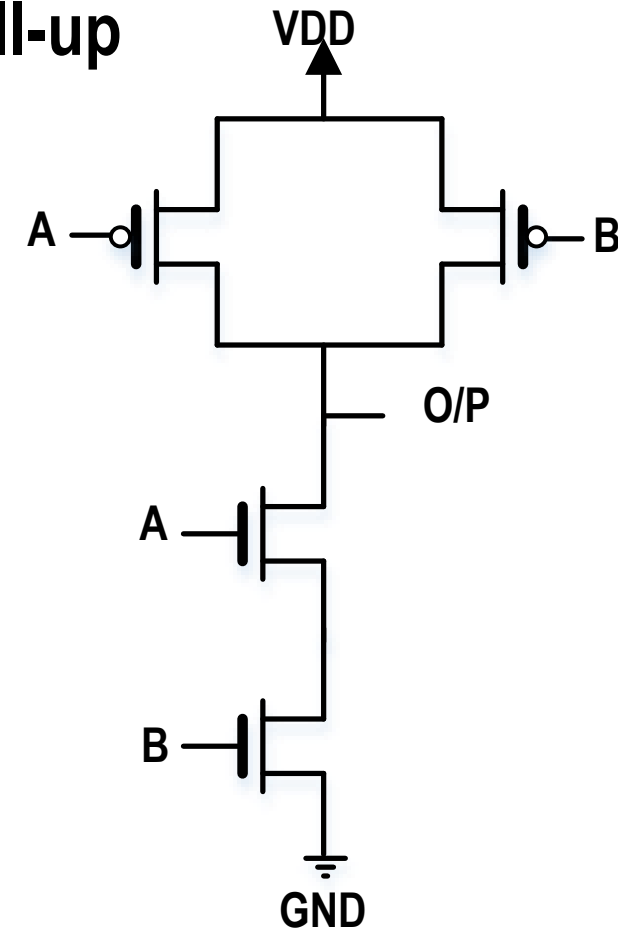


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Exercise_02



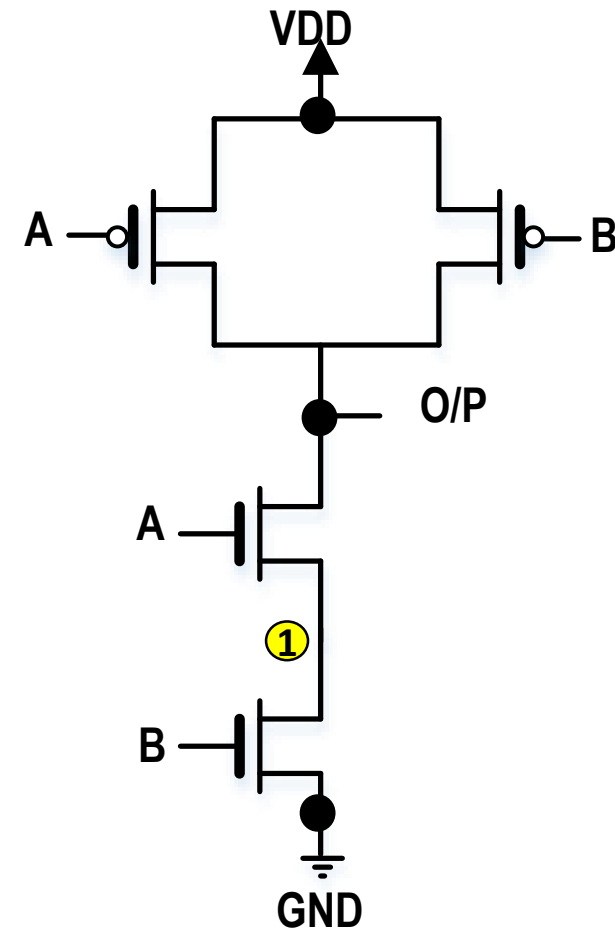
- Draw the stick diagram of the following circuit (NAND)
- Steps1:- construct the logic diagram of the given function (pull-up and pull-down networks) → **given**



Exercise_02



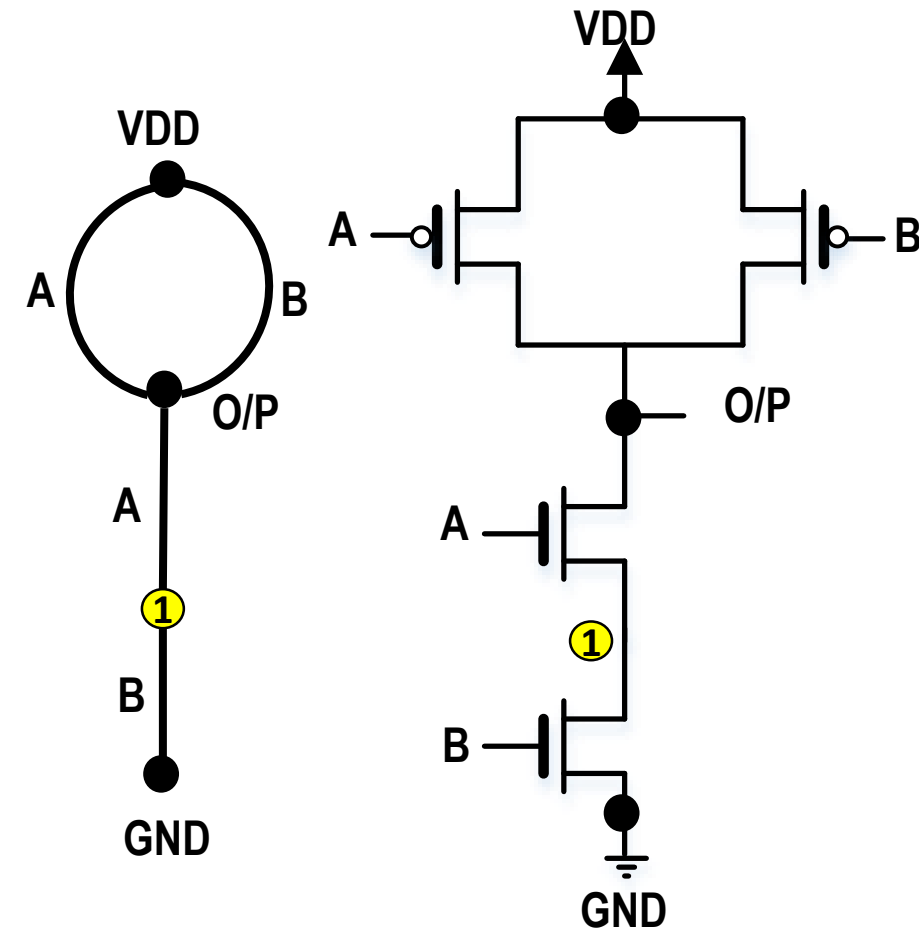
- ☐ Draw the stick diagram of the following circuit (NAND)
- ☐ Step2:- Identify each connection between transistors only and give them a unique name.
 - ☐ Main Nodes: VDD, O/P, GND
 - ☐ Sub-Nodes: 1



Exercise_02



- Draw the stick diagram of the following circuit (NAND)
- Step3:- Draw the Euler diagram



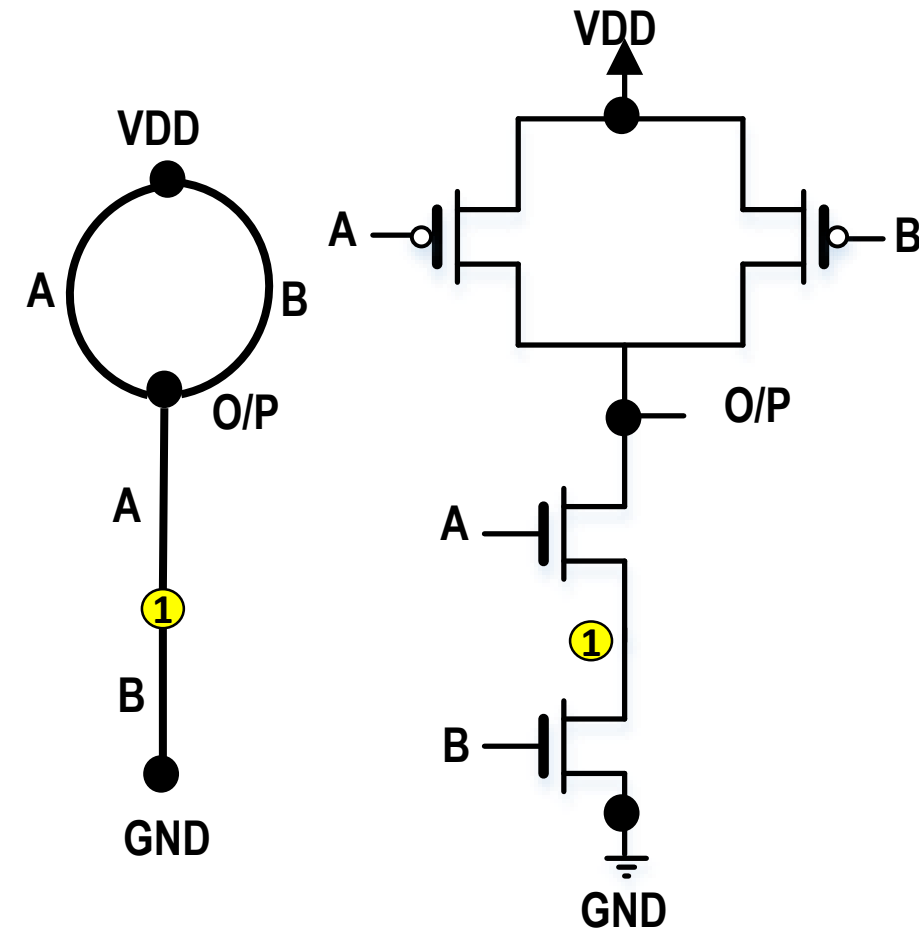
Exercise_02



- Draw the stick diagram of the following circuit (NAND)
- Step4:- Identify the Euler path

Condition of Euler Path

1. Nodes can be repeated but Paths cannot be repeated.
2. The path should be valid in both Pull-up and pull-down networks.



Exersice_02

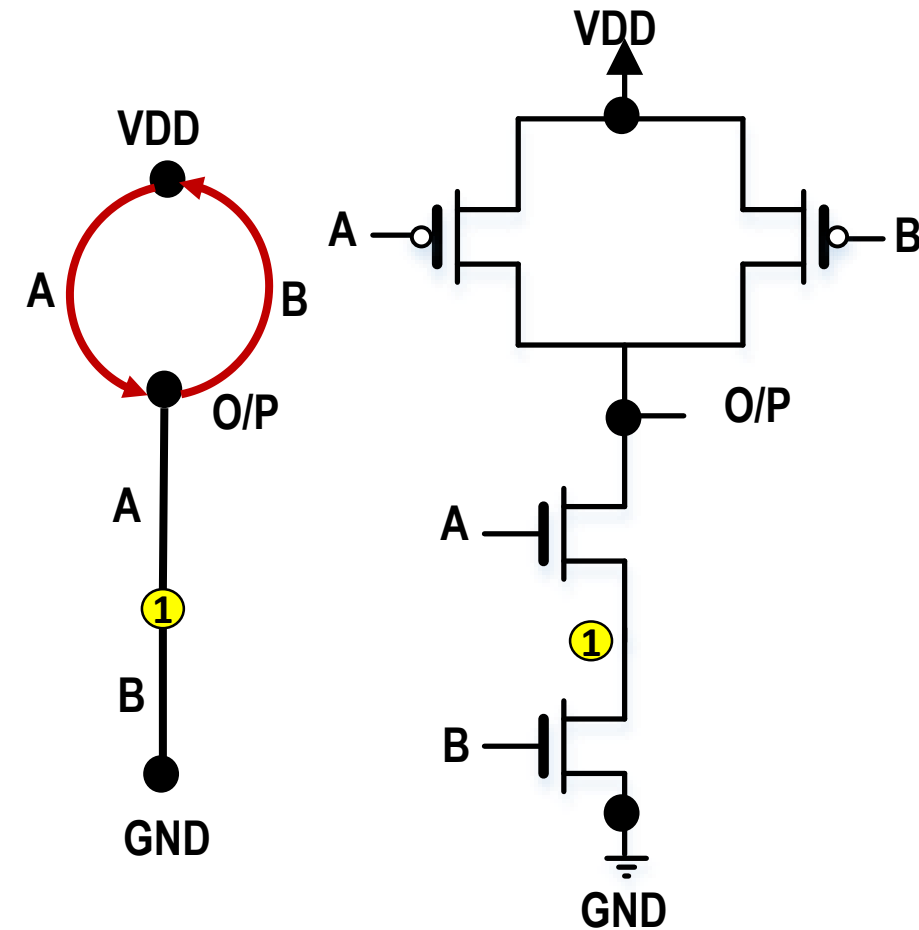


□ Draw the stick diagram of the following circuit (NAND)

□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→B



Exercise_02



□ Draw the stick diagram of the following circuit (NAND)

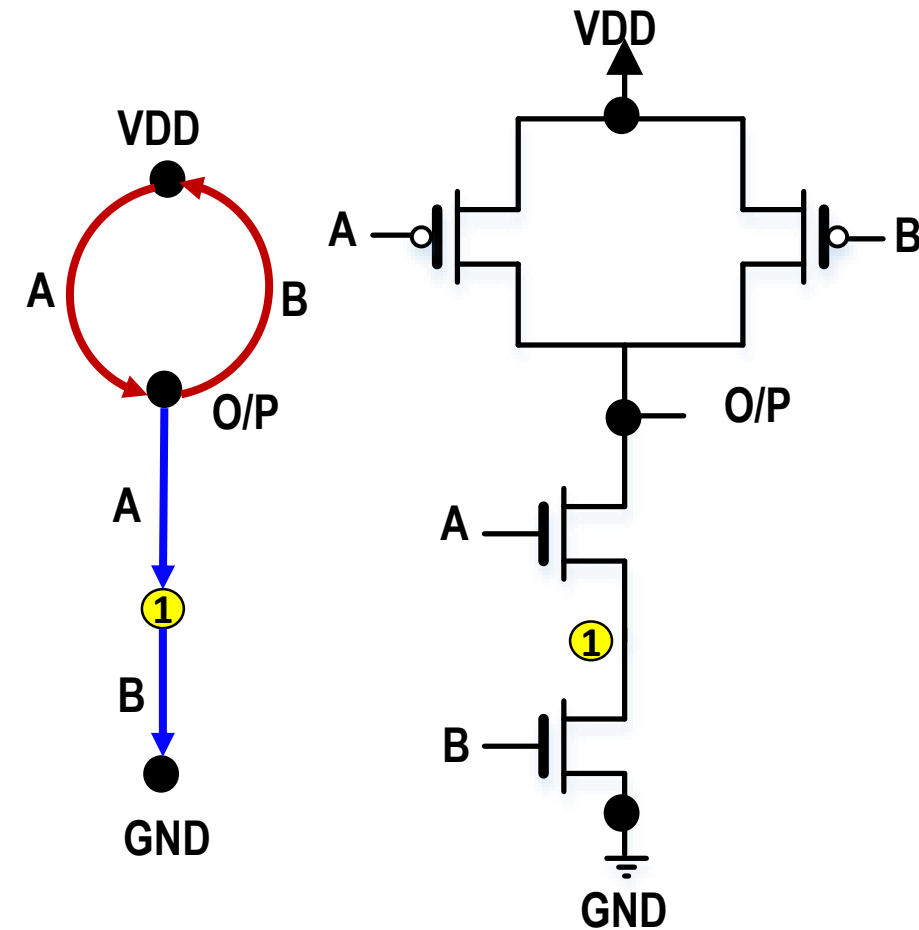
□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→B

Euler path for Pull-down network is

A→B



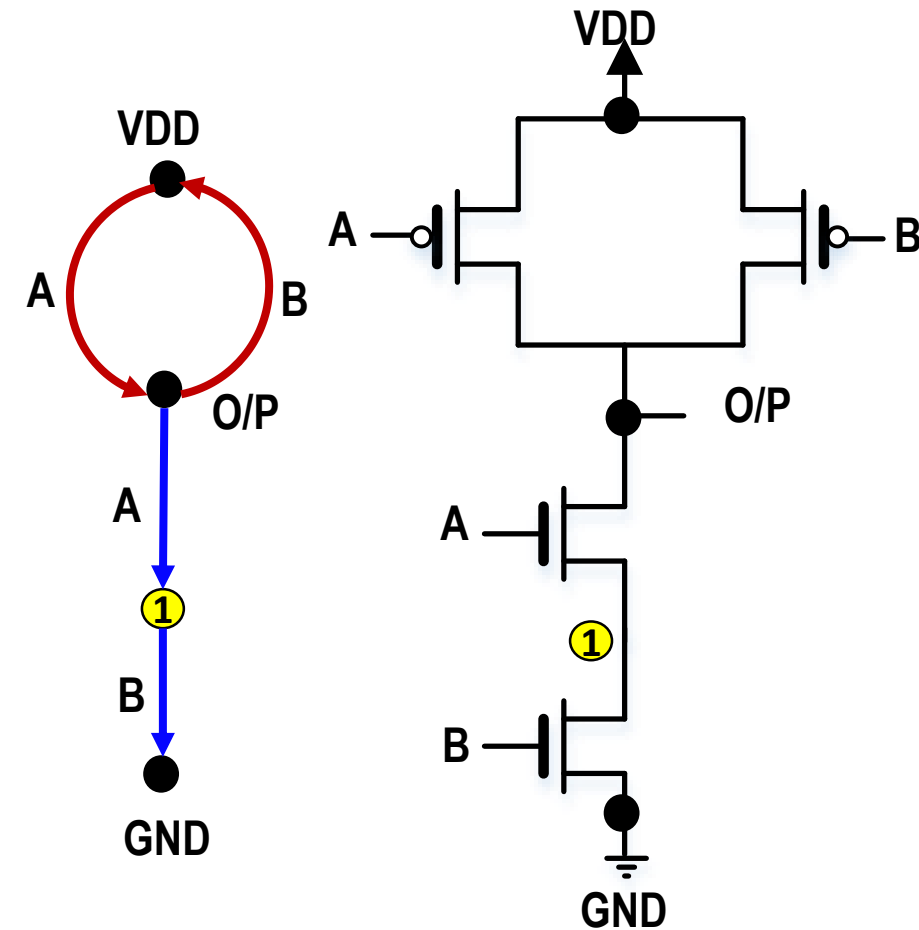
Exercise_02



□ Draw the stick diagram of the following circuit (NAND)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



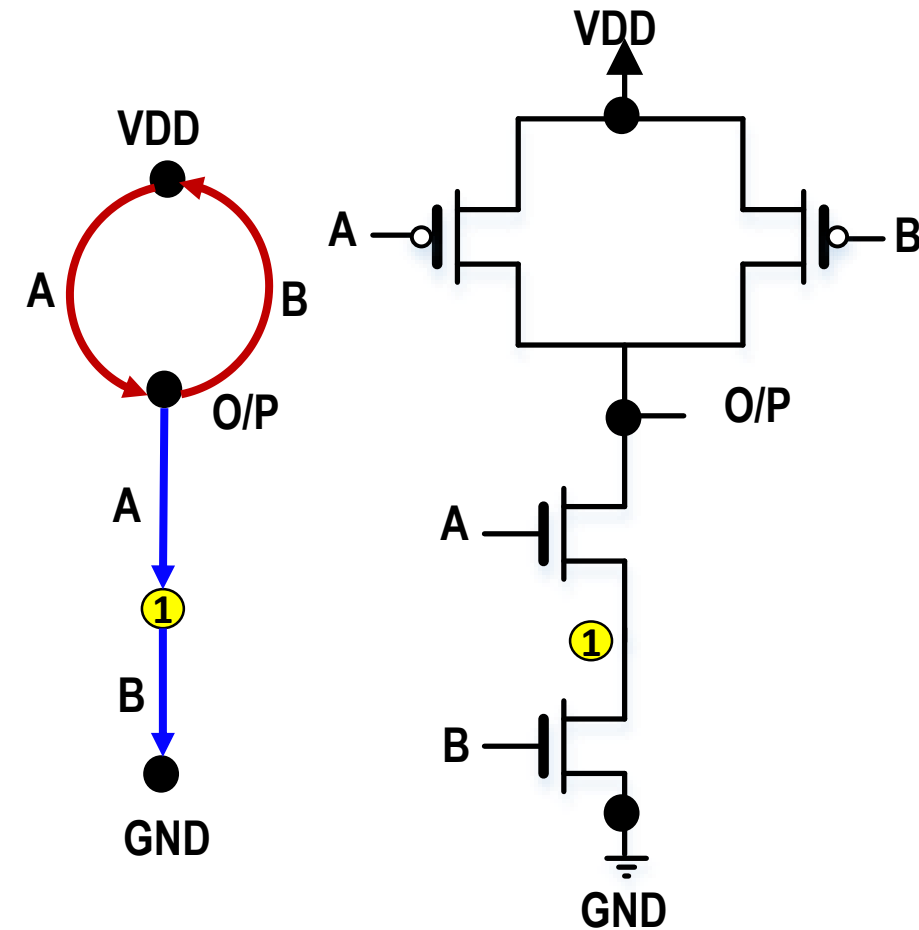
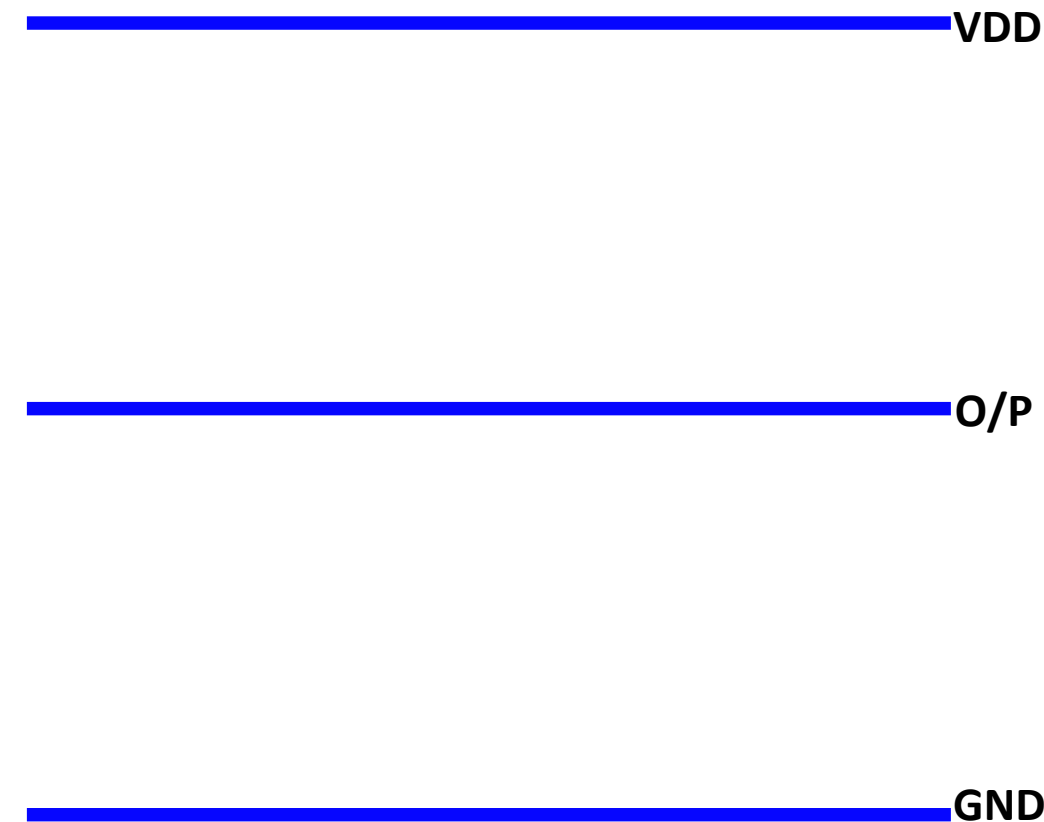
Exercise_02



□ Draw the stick diagram of the following circuit (NAND)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



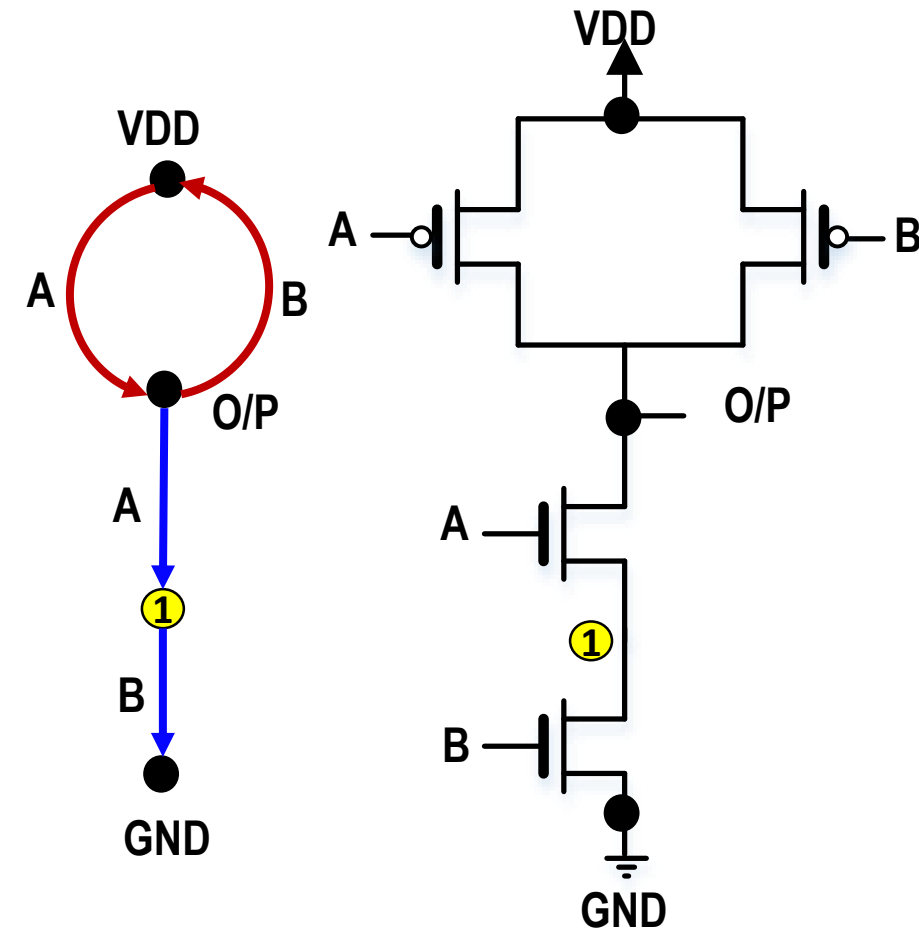
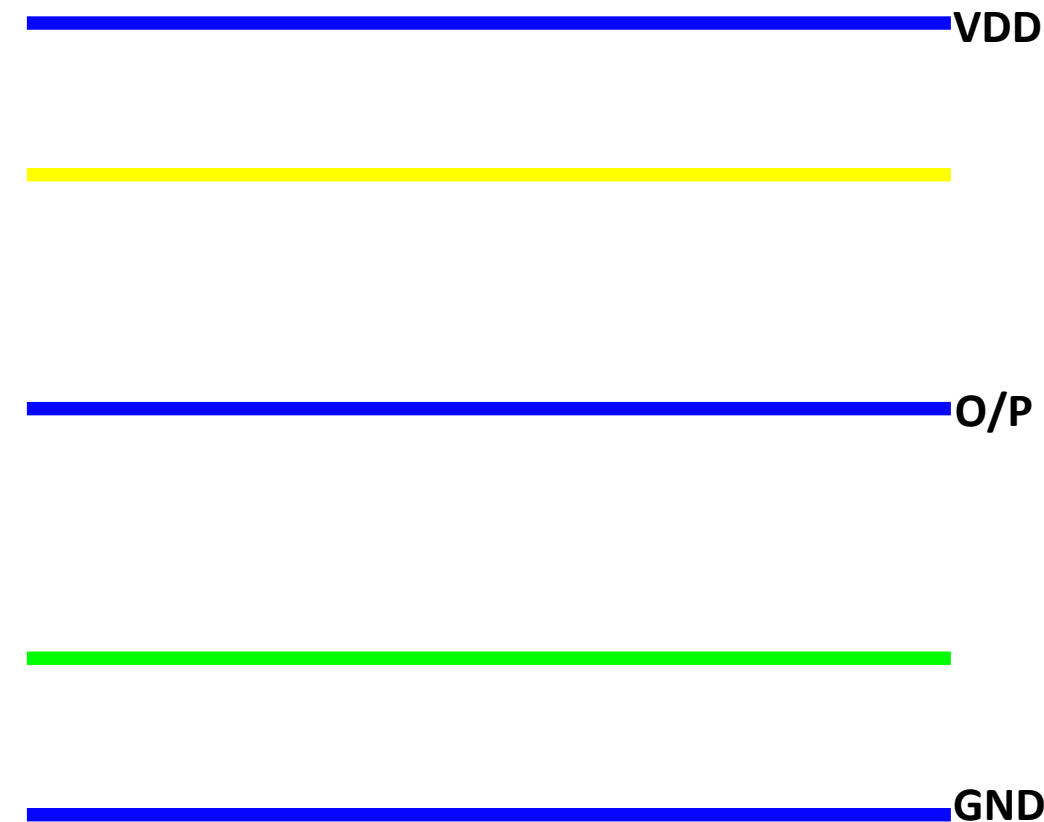
Exercise_02



☐ Draw the stick diagram of the following circuit (NAND)

☐ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



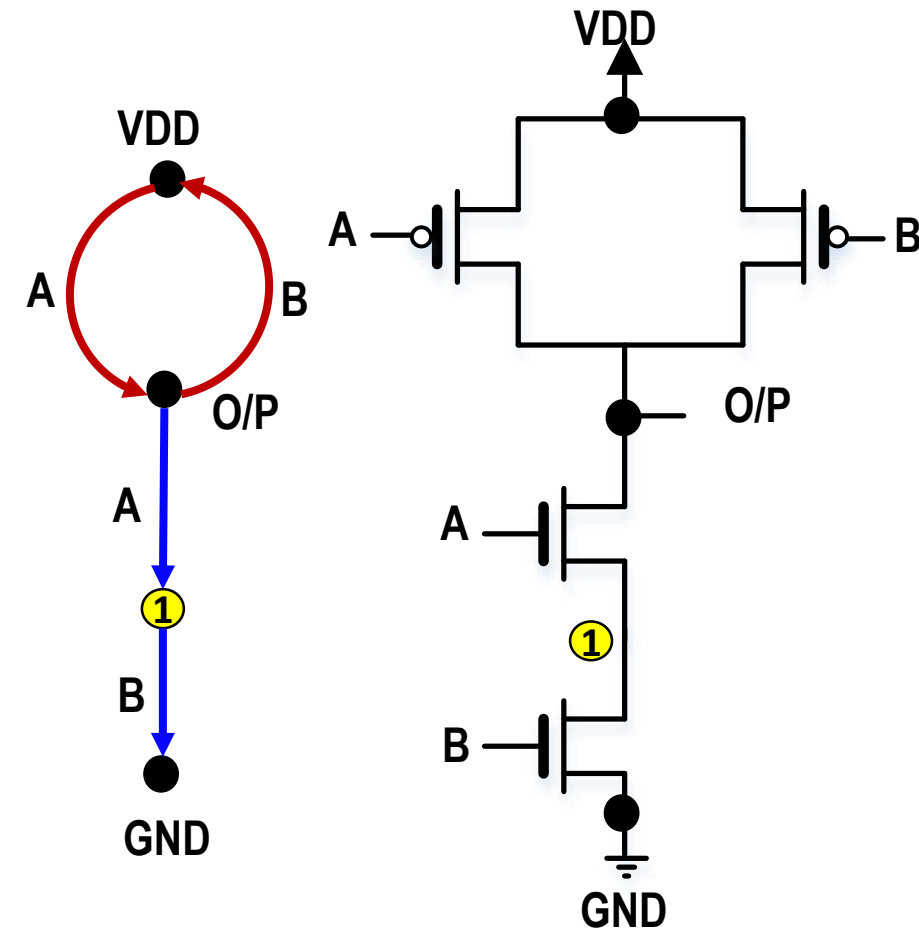
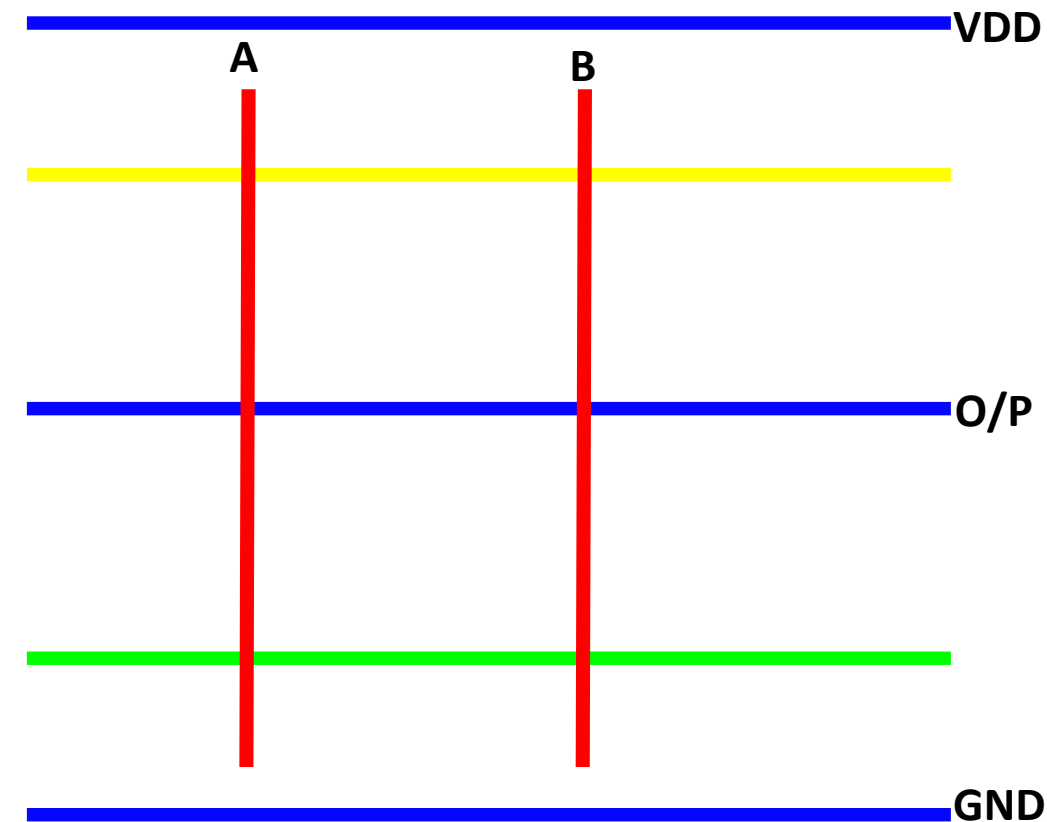
Exersice_02



☐ Draw the stick diagram of the following circuit (NAND)

☐ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



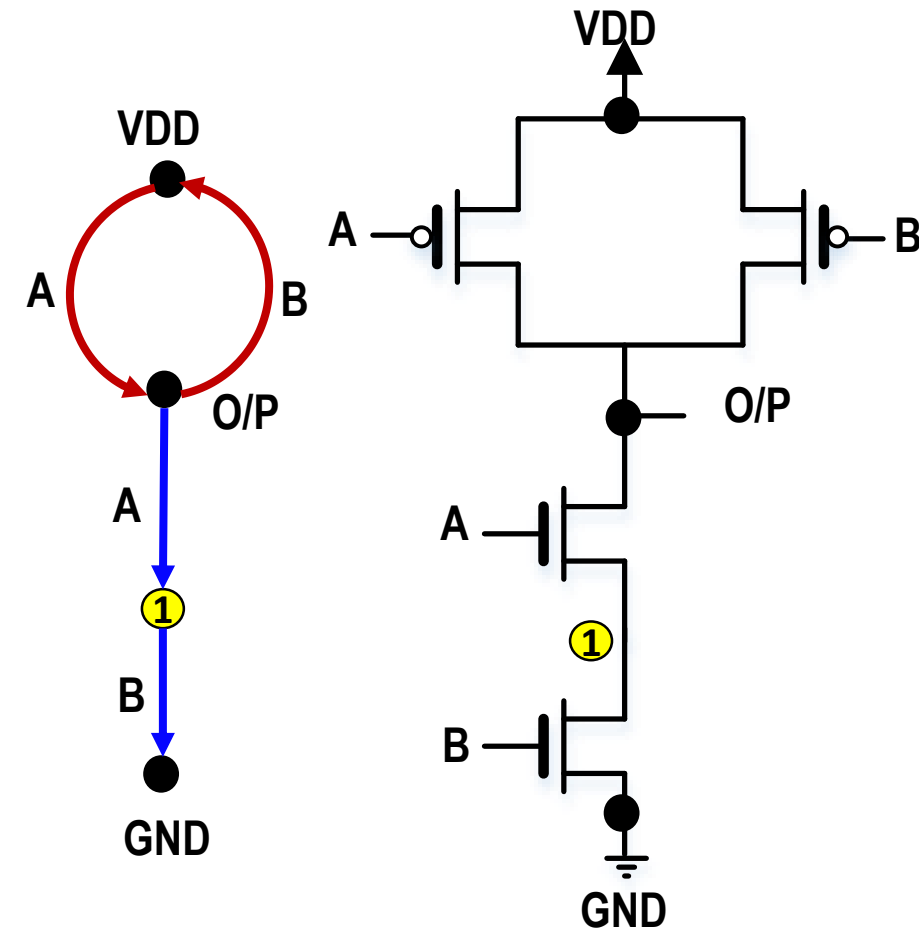
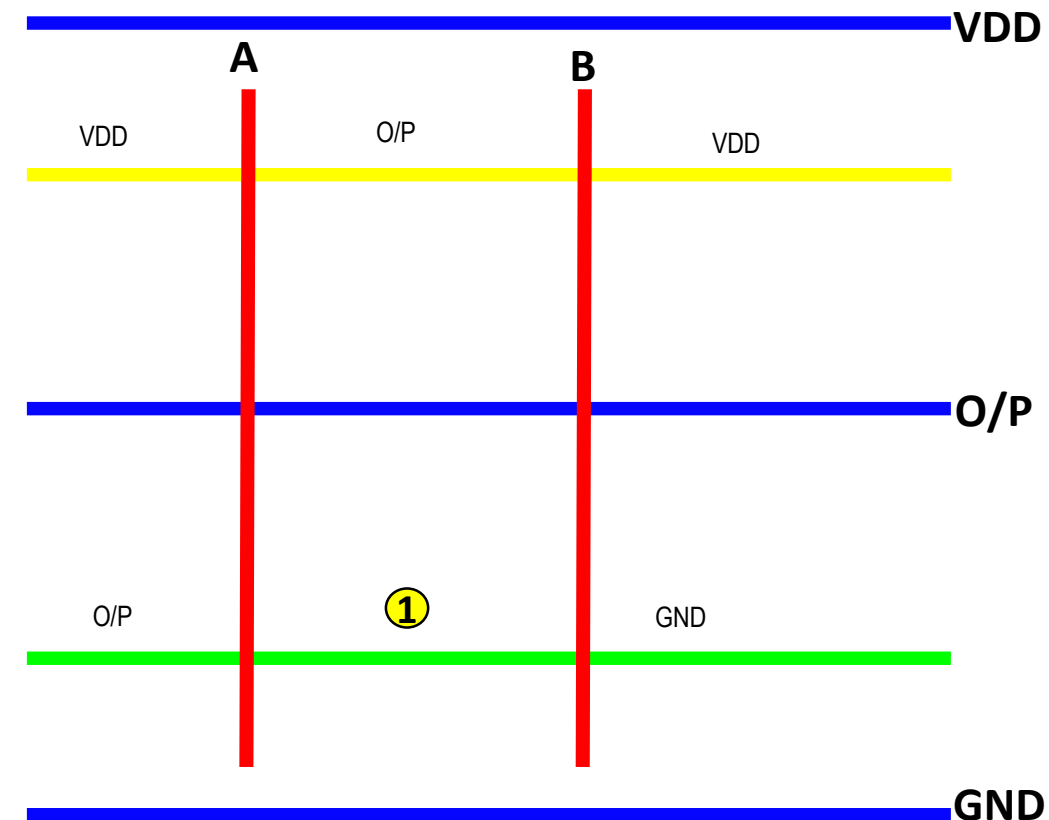
Exercise_02



□ Draw the stick diagram of the following circuit (NAND)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



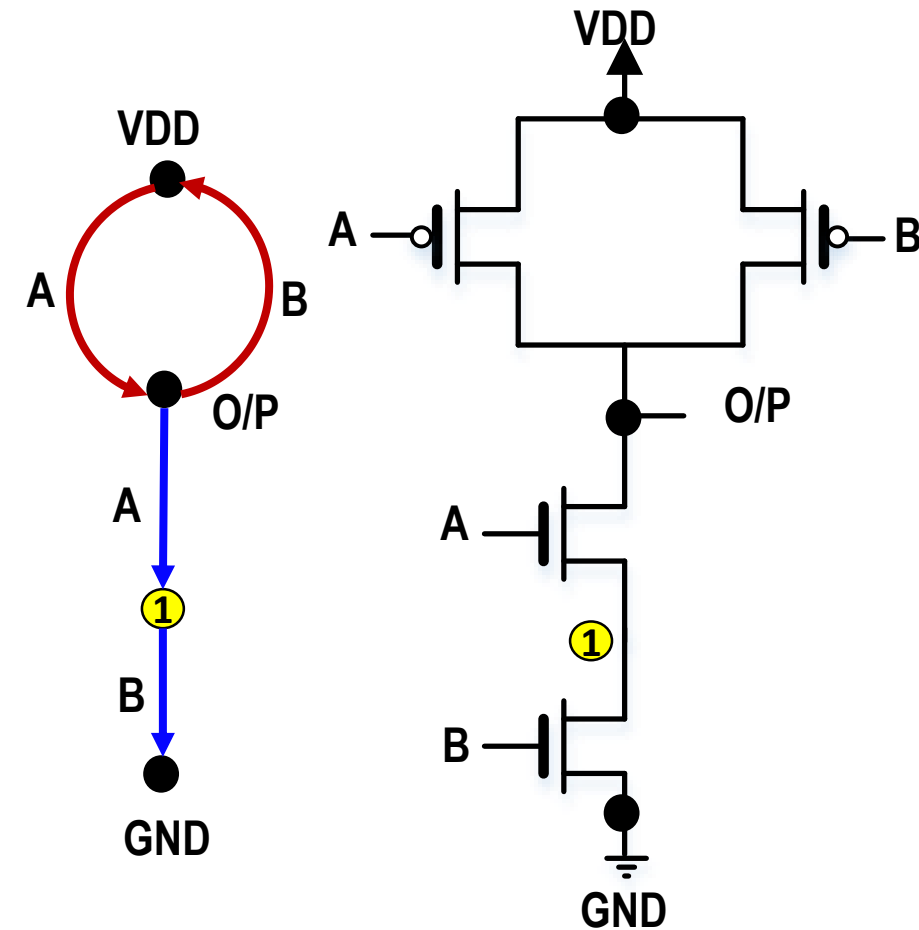
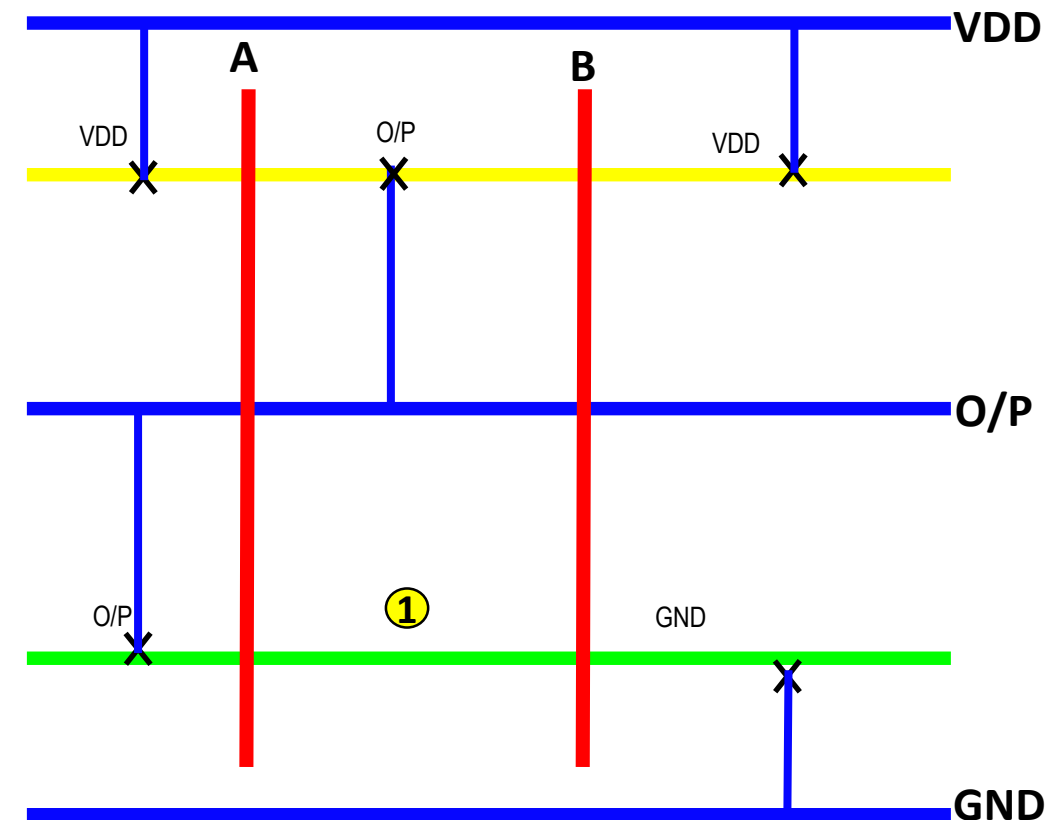
Exercise_02



□ Draw the stick diagram of the following circuit (NAND)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



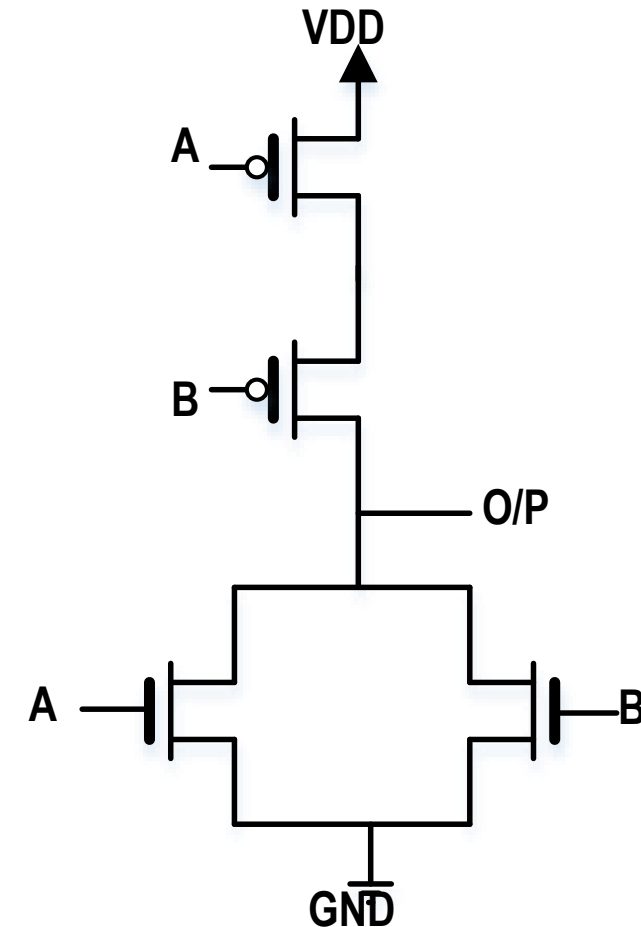


- ☐ Living in a 3D World
- ☐ Design Rules Check
- ☐ Geometric Design Rules
- ☐ Layout Methodologies
- ☐ Stick diagram
- ☐ Exercise_01
- ☐ Exercise_02
- ☐ **Exercise_03**
- ☐ Exercise_04

Exersice_03



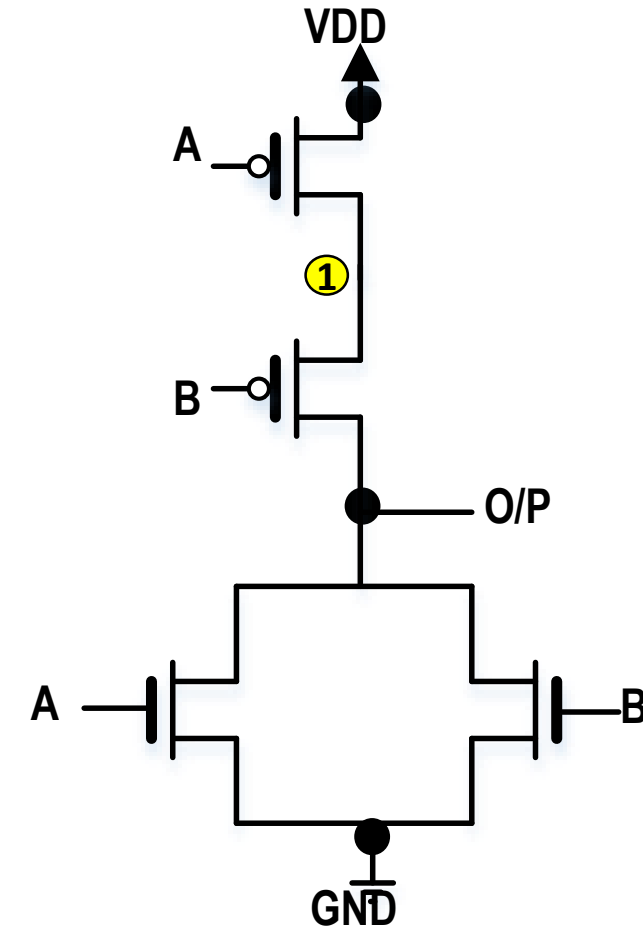
- Draw the stick diagram of the following circuit (NOR)
- Steps1:- construct the logic diagram of the given function (pull-up and pull-down networks) → **given**



Exercise_03



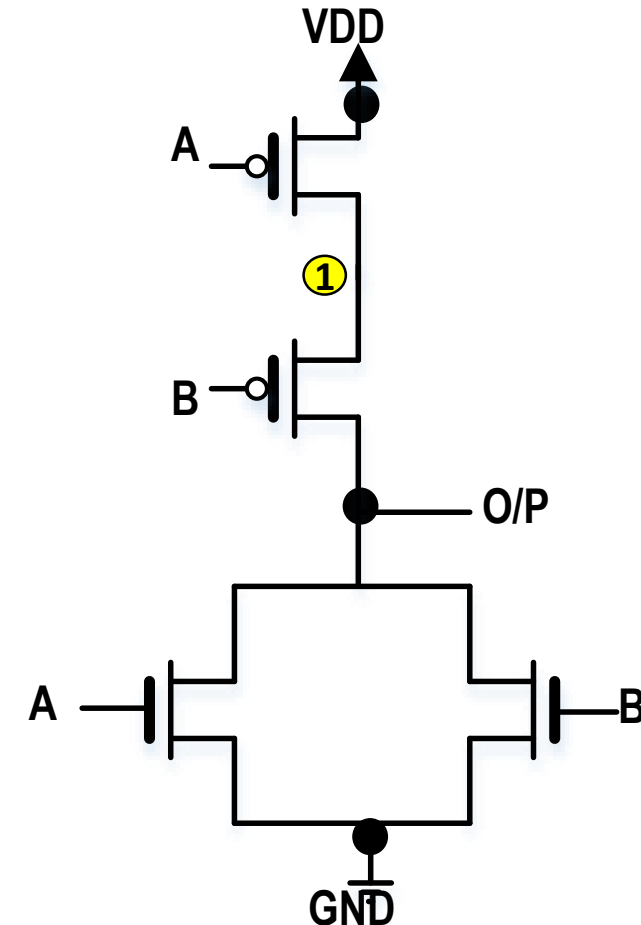
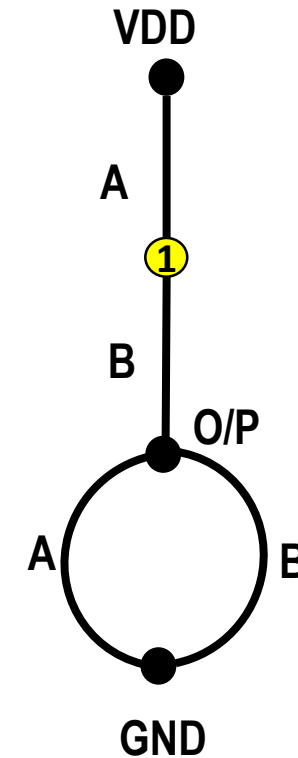
- ☐ Draw the stick diagram of the following circuit (NOR)
- ☐ Step2:- Identify each connection between transistors only and give them a unique name.
 - ☐ Main Nodes: VDD, O/P, GND
 - ☐ Sub-Nodes: 1



Exersice_03



- Draw the stick diagram of the following circuit (NOR)
- Step3:- Draw the Euler diagram



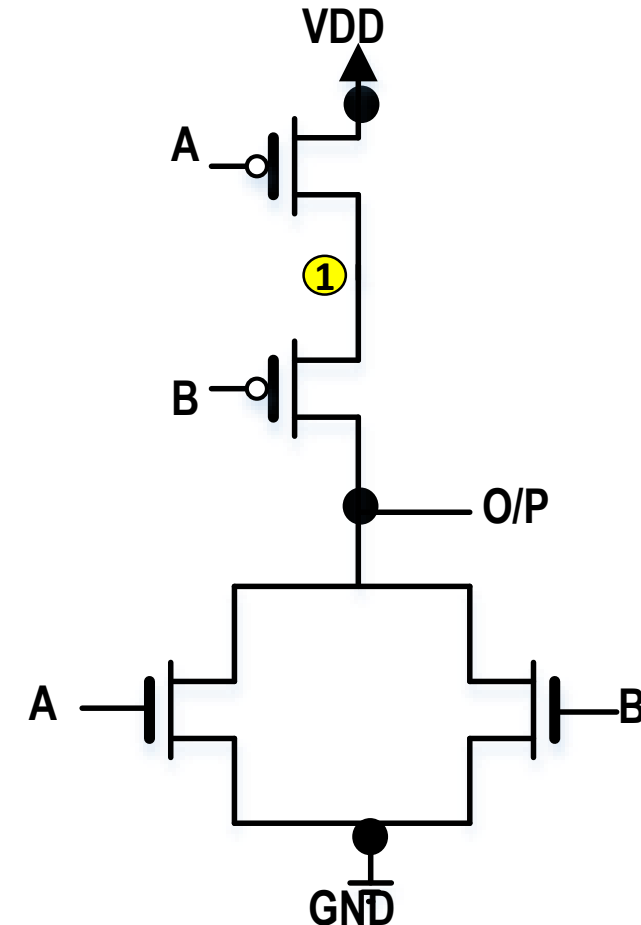
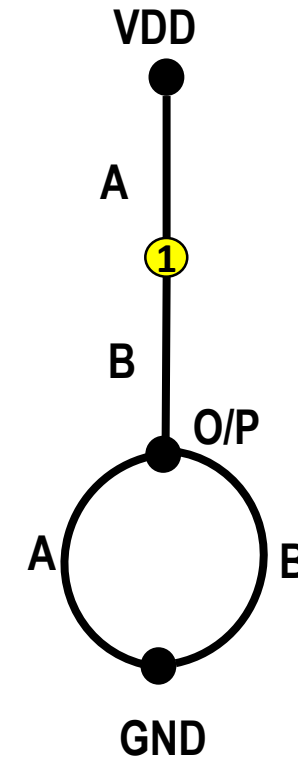
Exercise_03



- Draw the stick diagram of the following circuit (NOR)
- Step4:- Identify the Euler path

Condition of Euler Path

1. Nodes can be repeated but Paths cannot be repeated.
2. The path should be valid in both Pull-up and pull-down networks.



Exercise_03

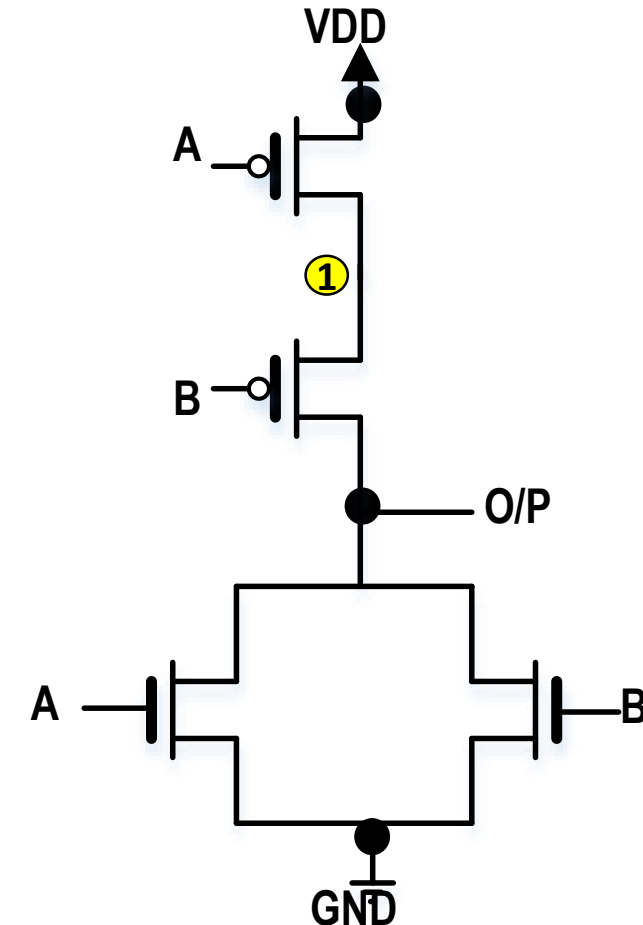
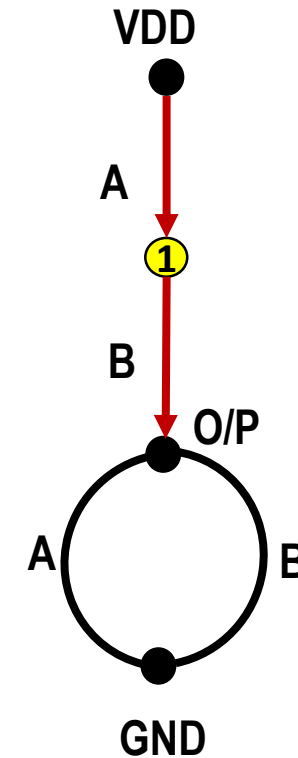


□ Draw the stick diagram of the following circuit (NOR)

□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→B



Exercise_03



□ Draw the stick diagram of the following circuit (NOR)

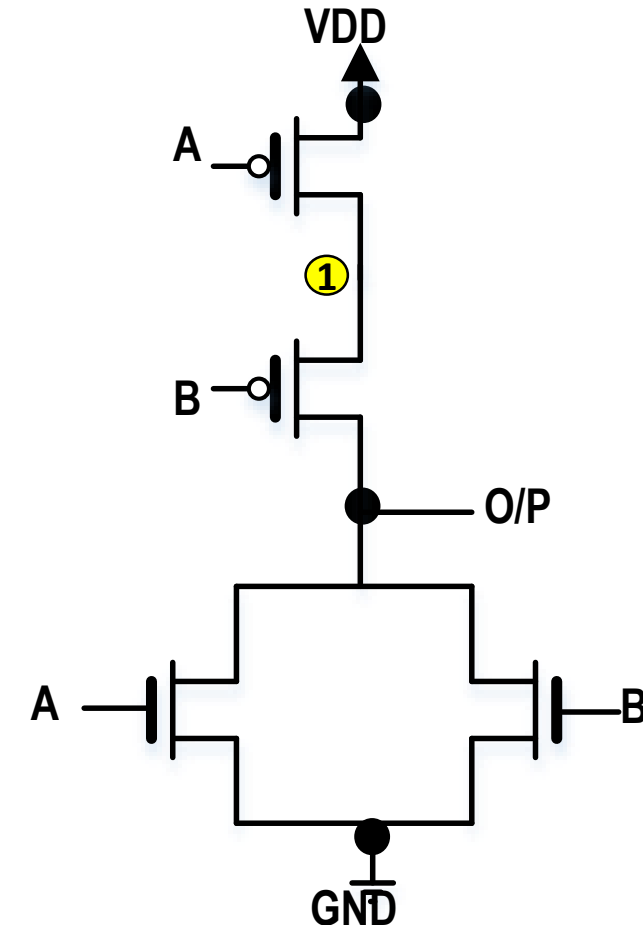
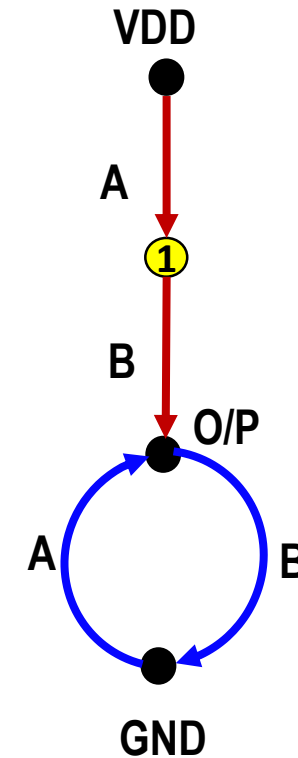
□ Step4:- Identify the Euler path

Euler path for Pull-up network is

A→B

Euler path for Pull-down network is

A→B



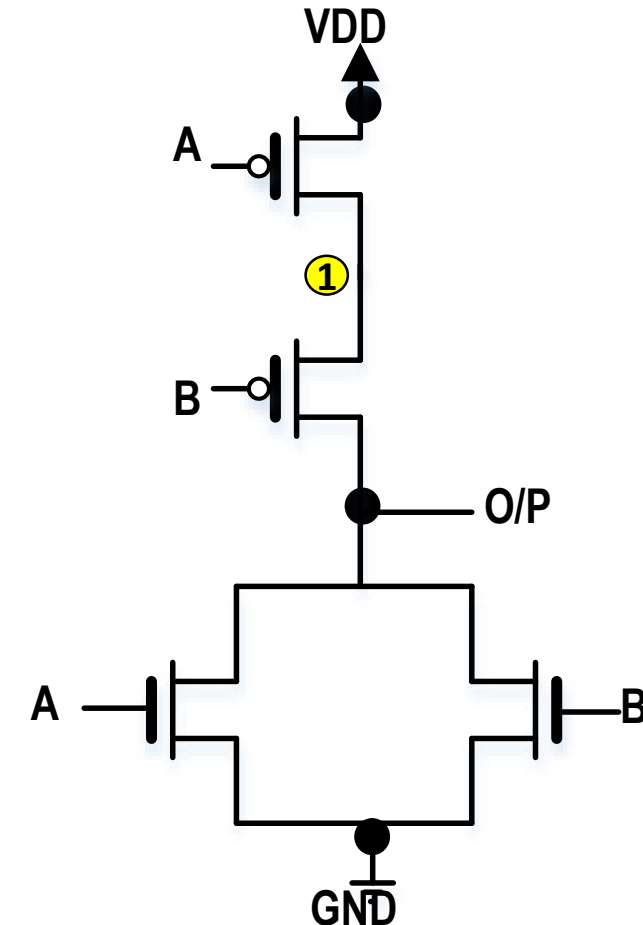
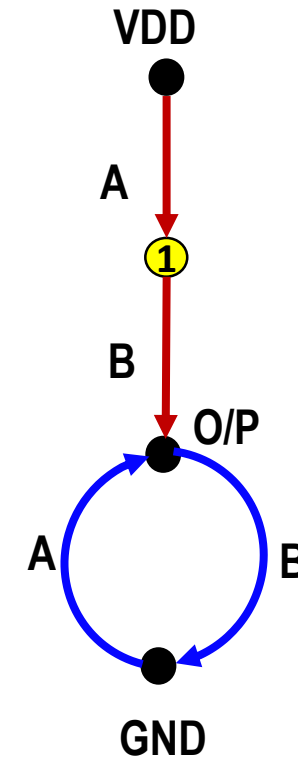
Exercise_03



□ Draw the stick diagram of the following circuit (NOR)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



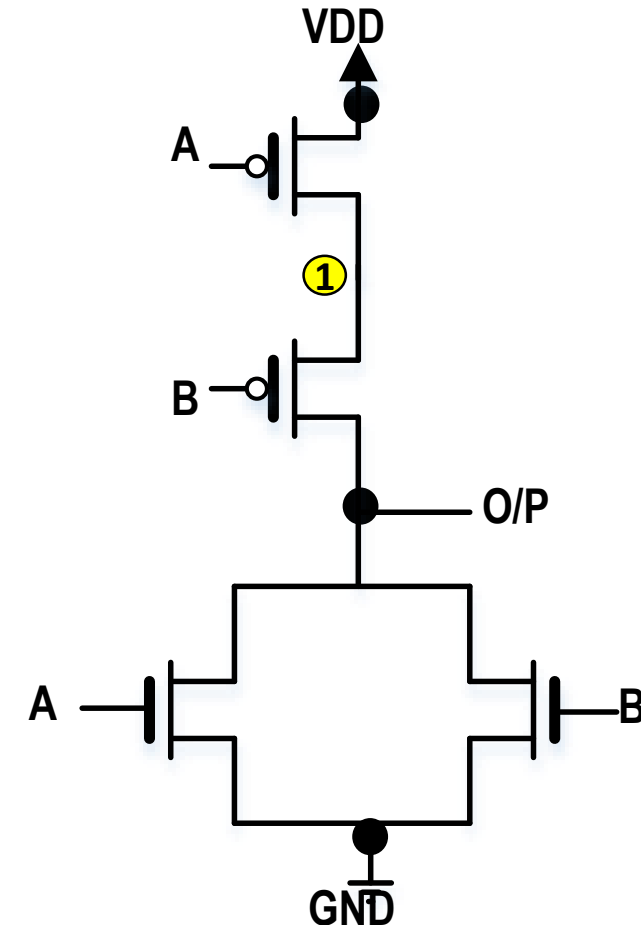
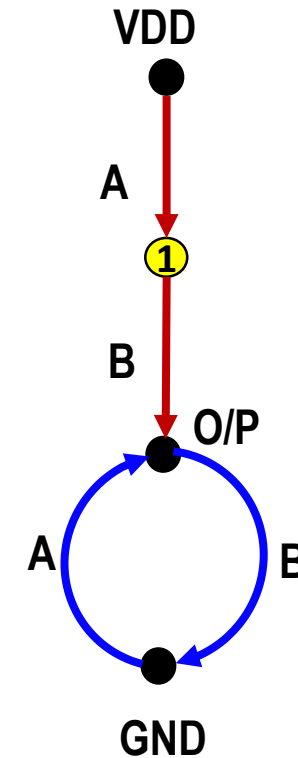
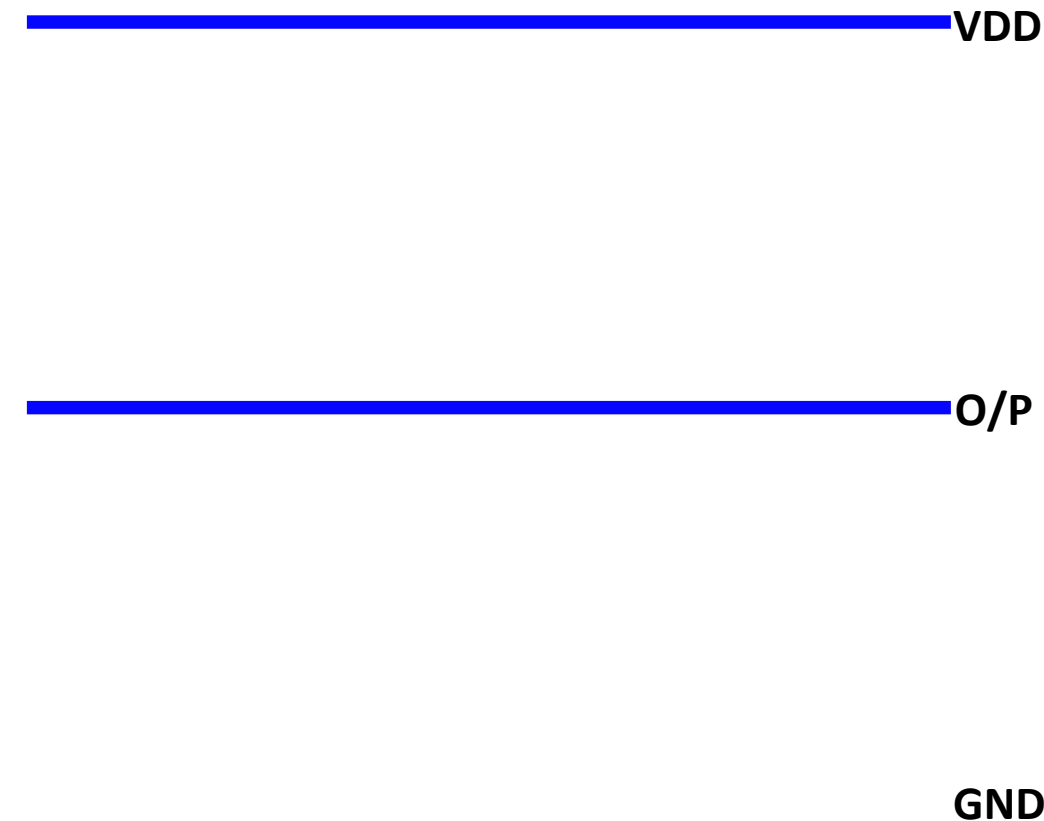
Exercise_03



□ Draw the stick diagram of the following circuit (NOR)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



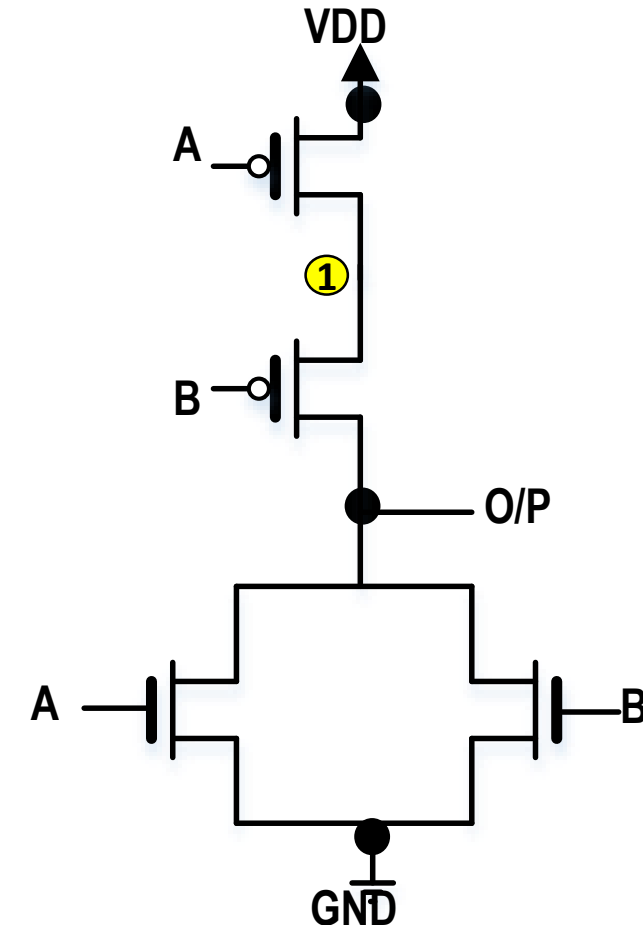
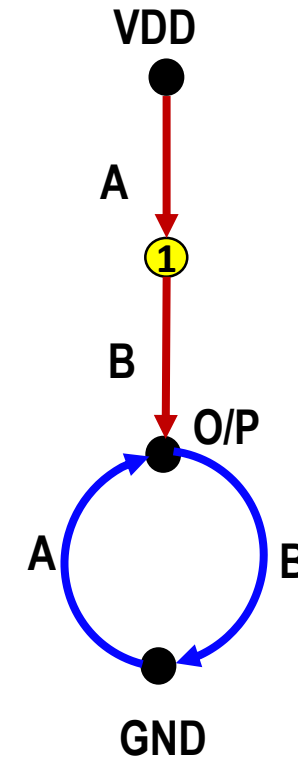
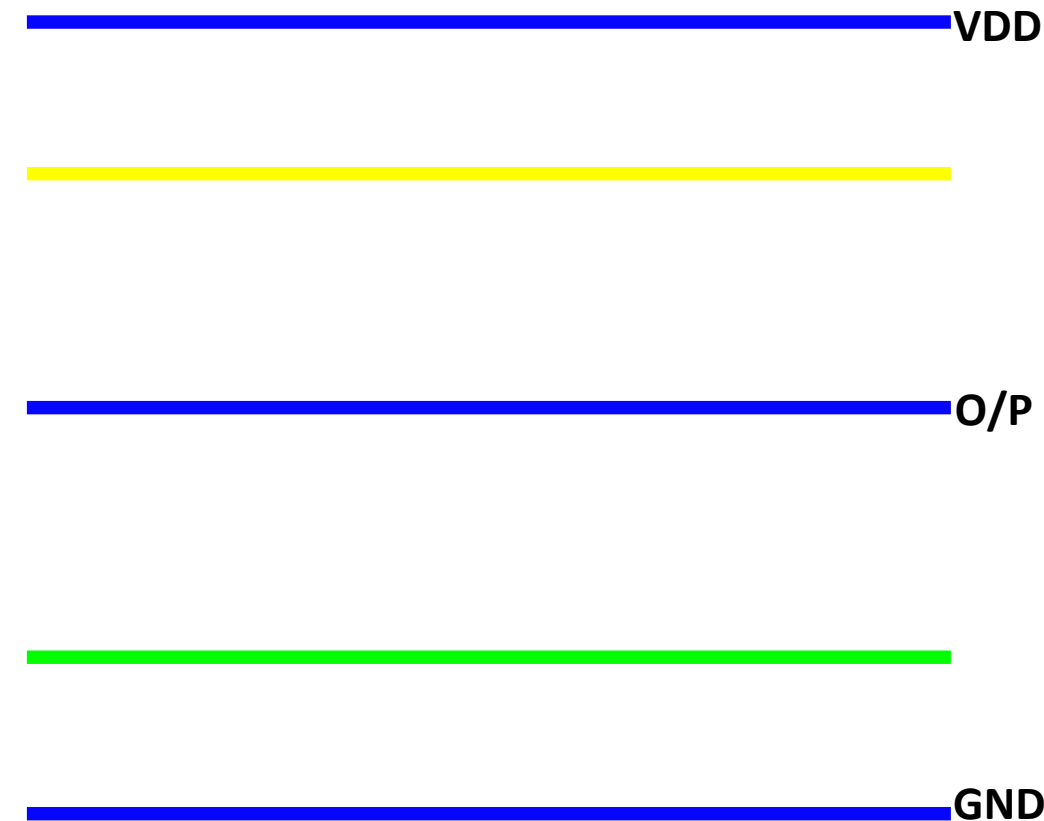
Exercise_03



☐ Draw the stick diagram of the following circuit (NOR)

☐ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



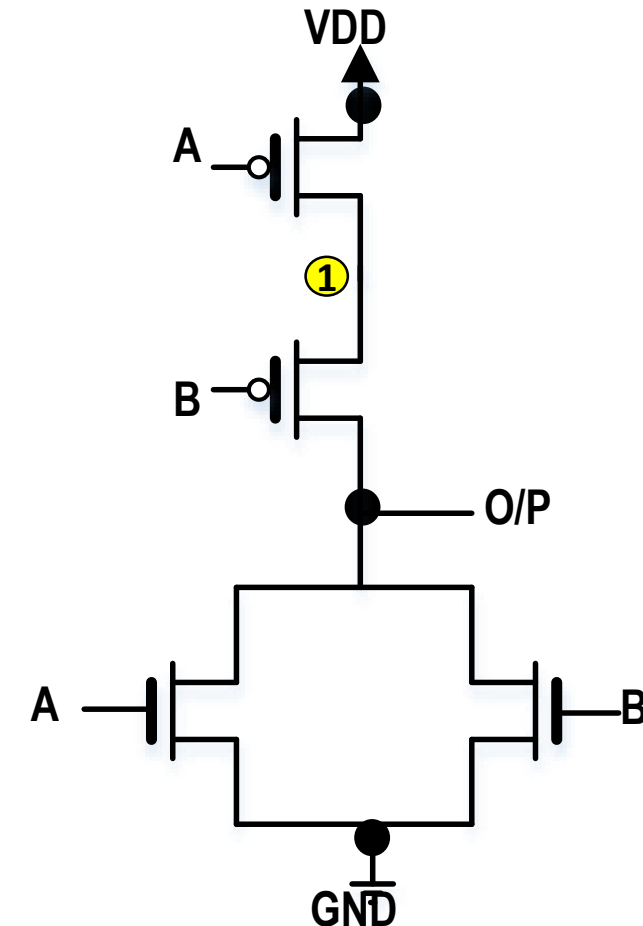
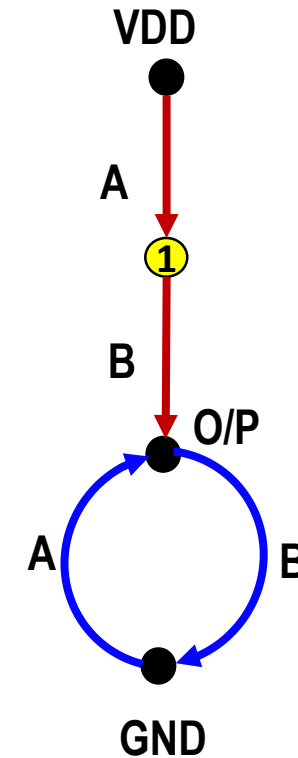
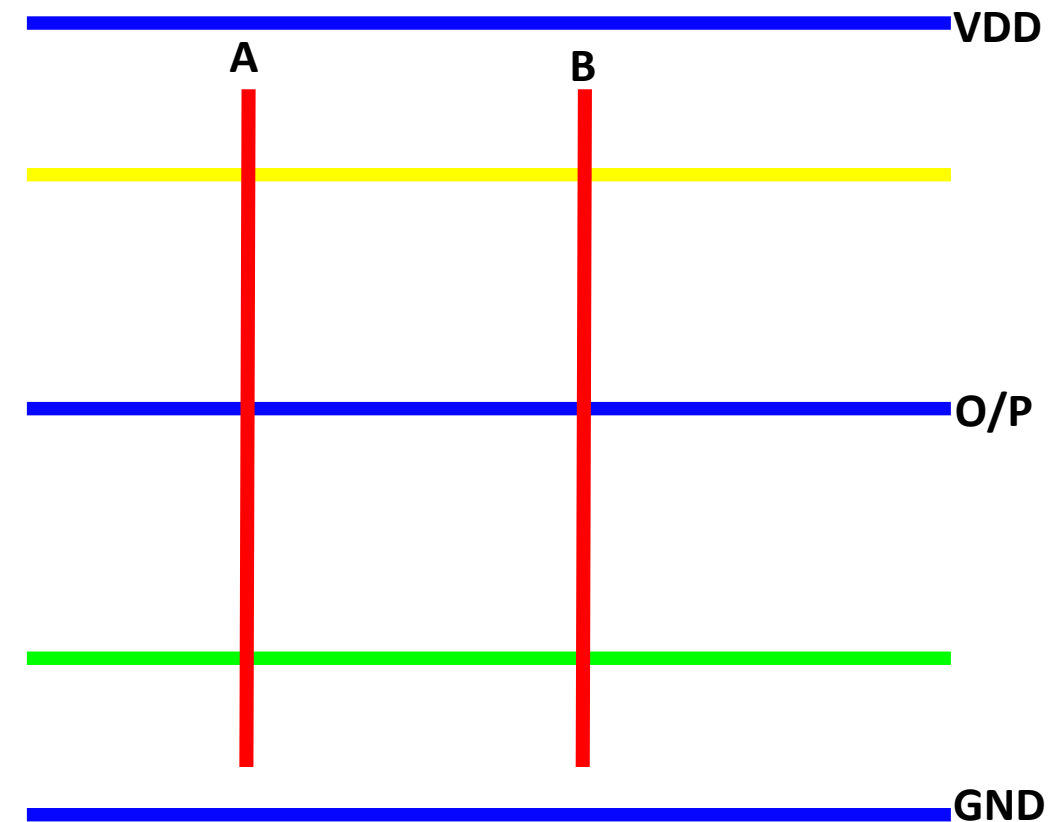
Exercise_03



□ Draw the stick diagram of the following circuit (NOR)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



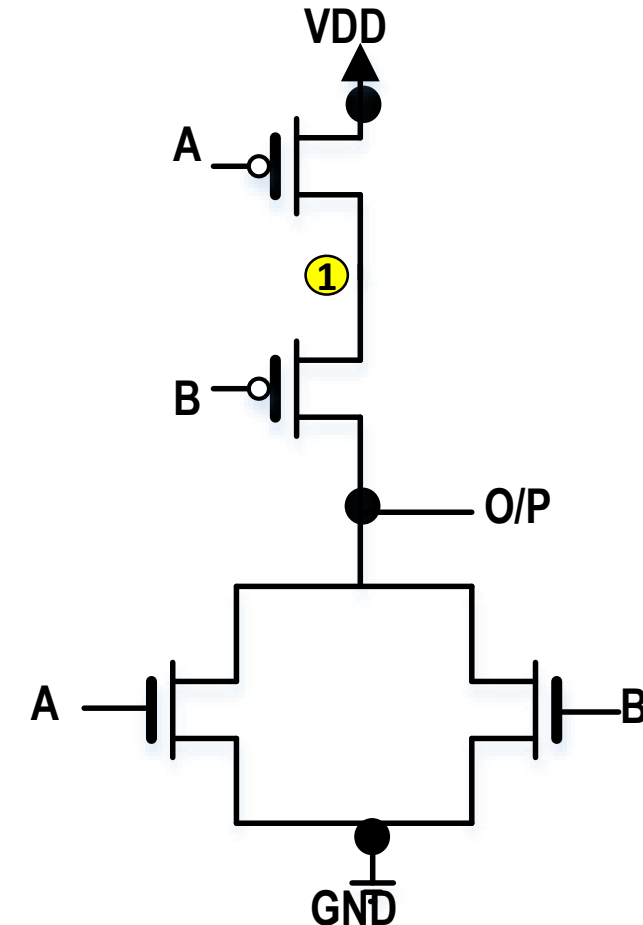
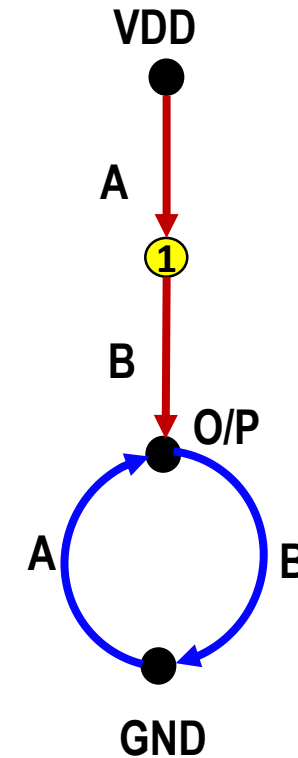
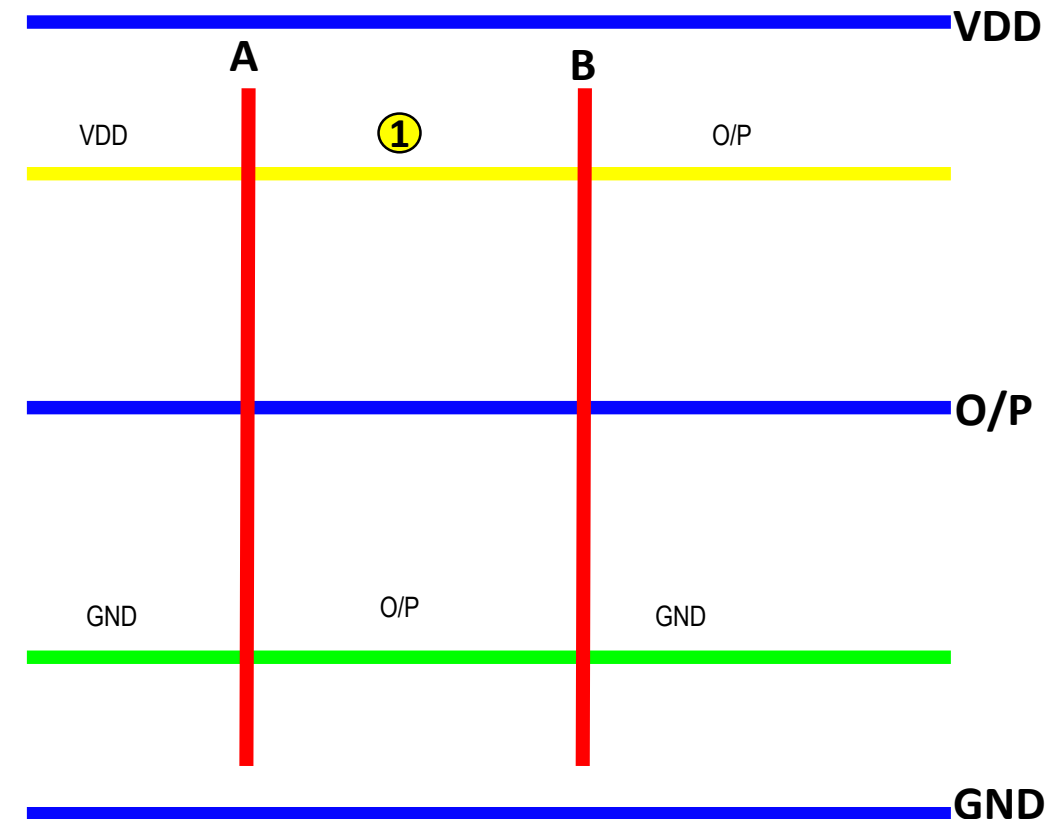
Exercise_03



□ Draw the stick diagram of the following circuit (NOR)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



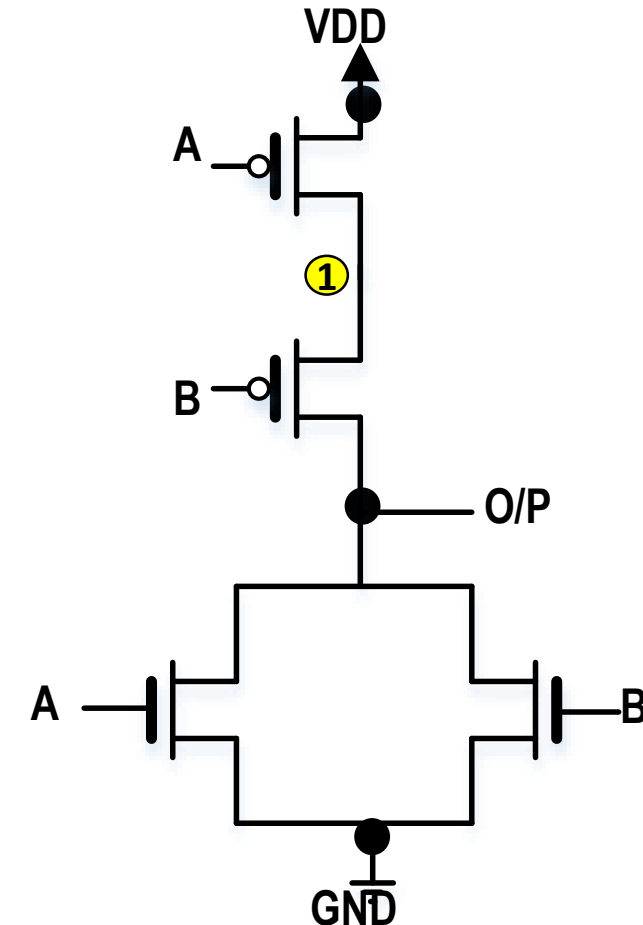
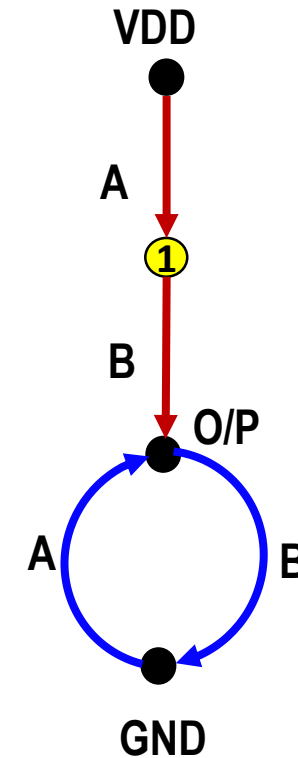
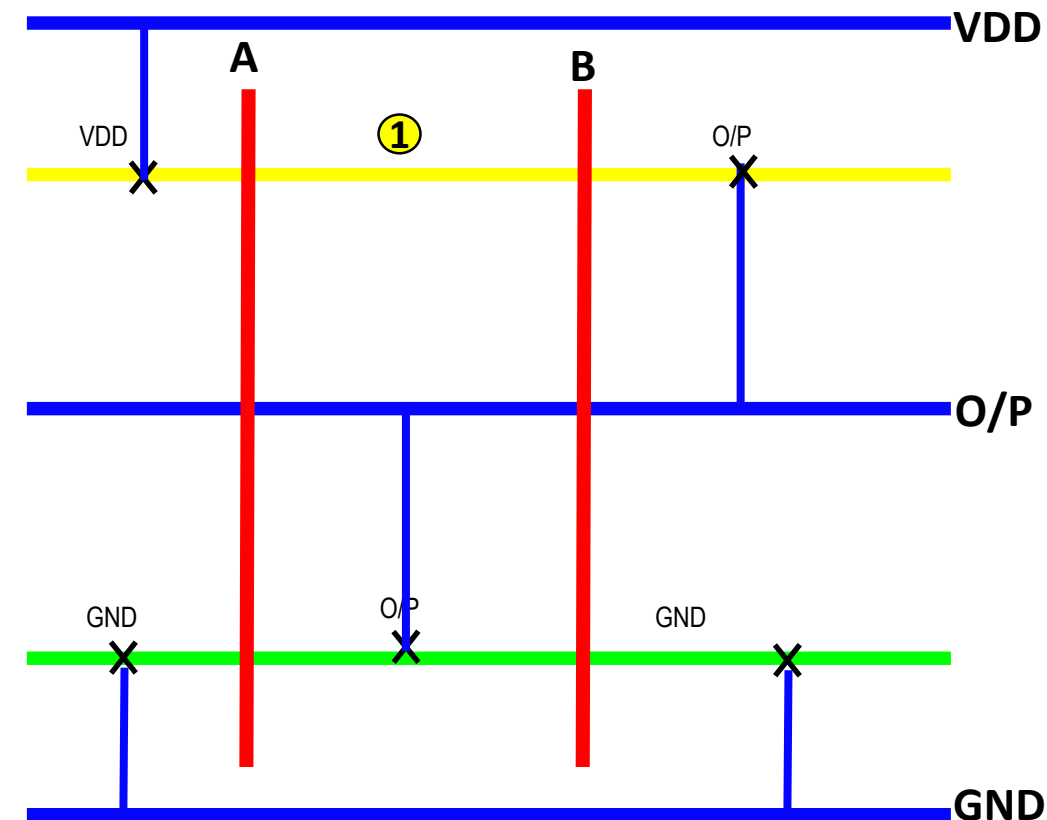
Exercise_03



□ Draw the stick diagram of the following circuit (NOR)

□ Step5:- Draw the stick diagram

$A \rightarrow B$ (in order)



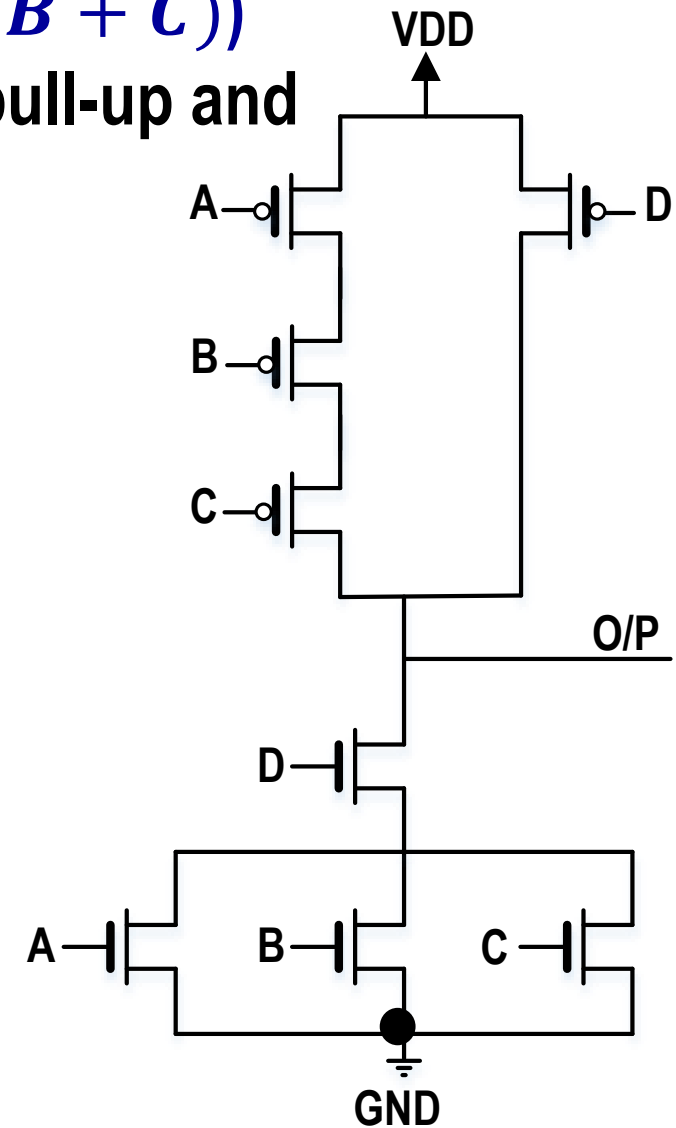


- ☐ Living in a 3D World
- ☐ Design Rules Check
- ☐ Geometric Design Rules
- ☐ Layout Methodologies
- ☐ Stick diagram
- ☐ Exercise_01
- ☐ Exercise_02
- ☐ Exercise_03
- ☐ **Exercise_04**

Exersice_04



- Draw the stick diagram of the following circuit ($f = \overline{D \cdot (A + B + C)}$)
- Step1:- construct the logic diagram of the given function (pull-up and pull-down networks) → **given**



Exercise_04

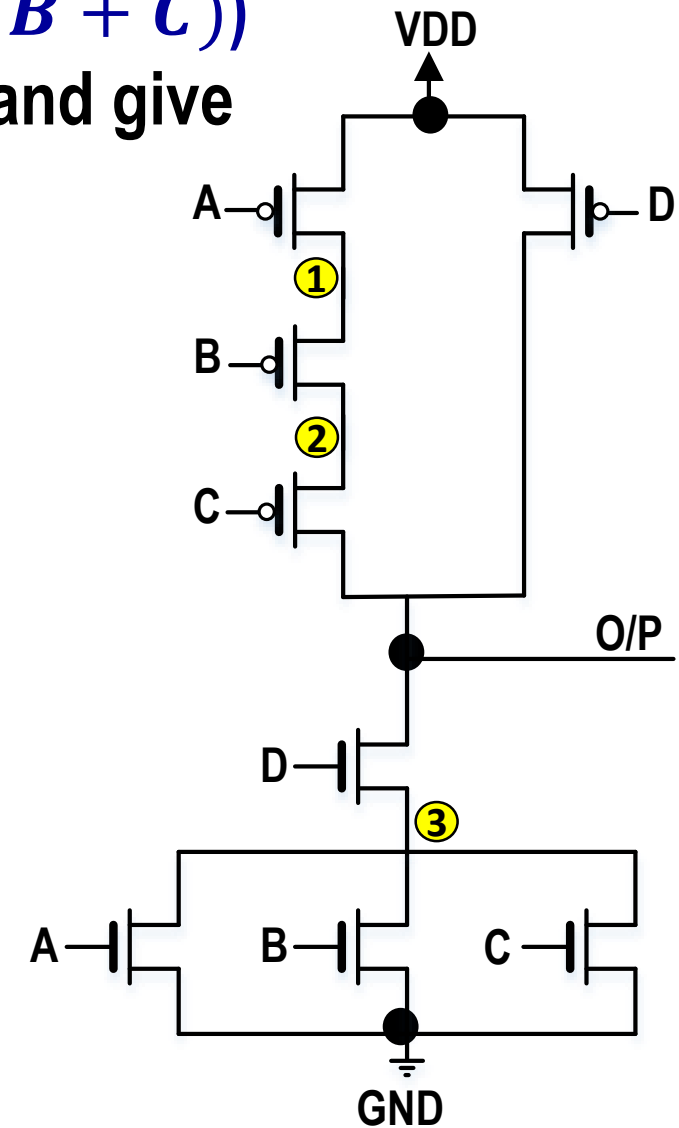


□ Draw the stick diagram of the following circuit ($f = \overline{D \cdot (A + B + C)}$)

□ Step2:- Identify each connection between transistors only and give them a unique name.

□ Main Nodes: VDD, O/P, GND

□ Sub-Nodes: 1,2,3

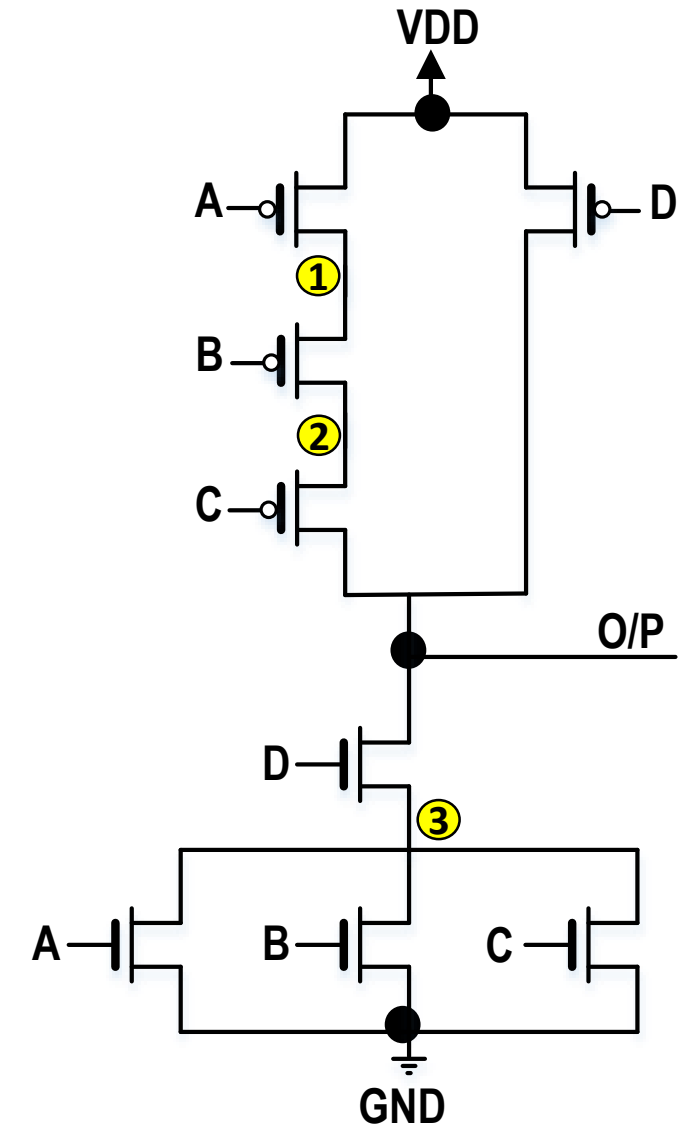
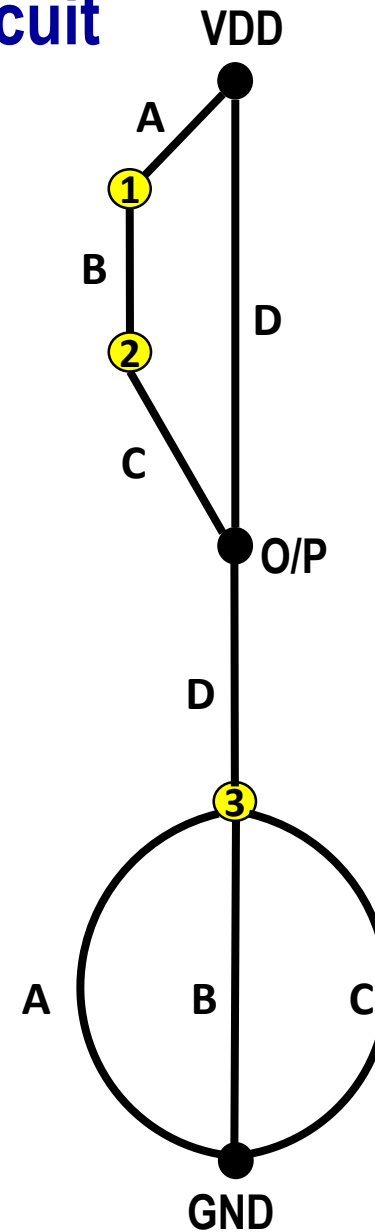


Exersice_04



□ Draw the stick diagram of the following circuit
($f = \overline{D \cdot (A + B + C)}$)

□ Step3:- Draw the Euler diagram



Exercise_04



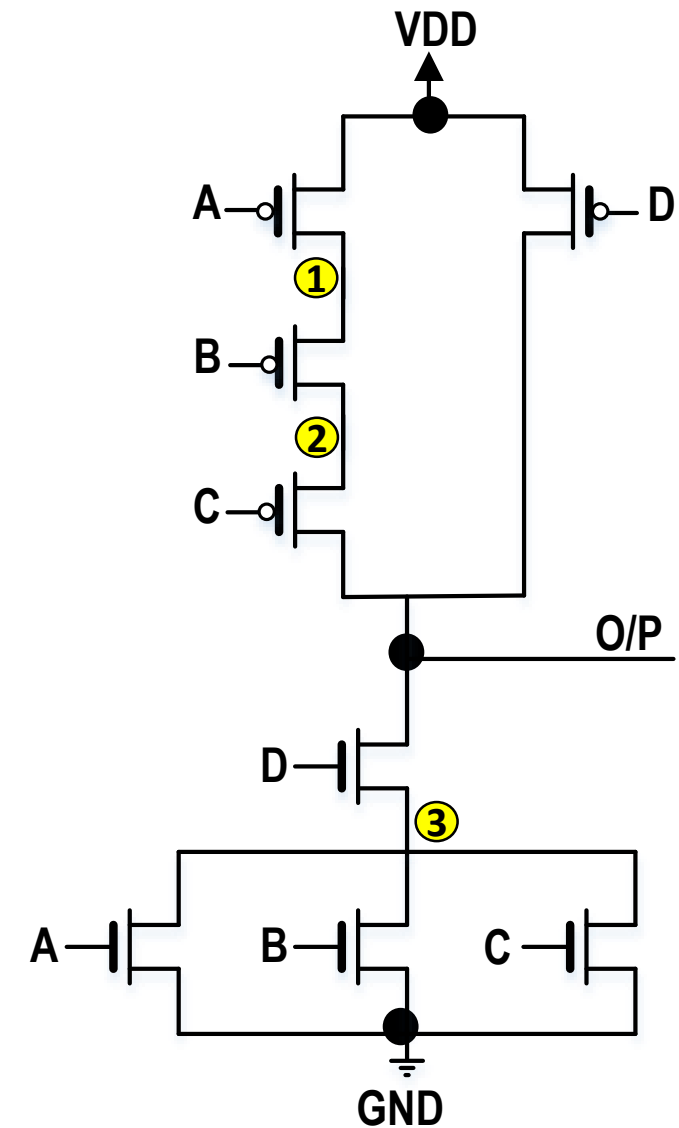
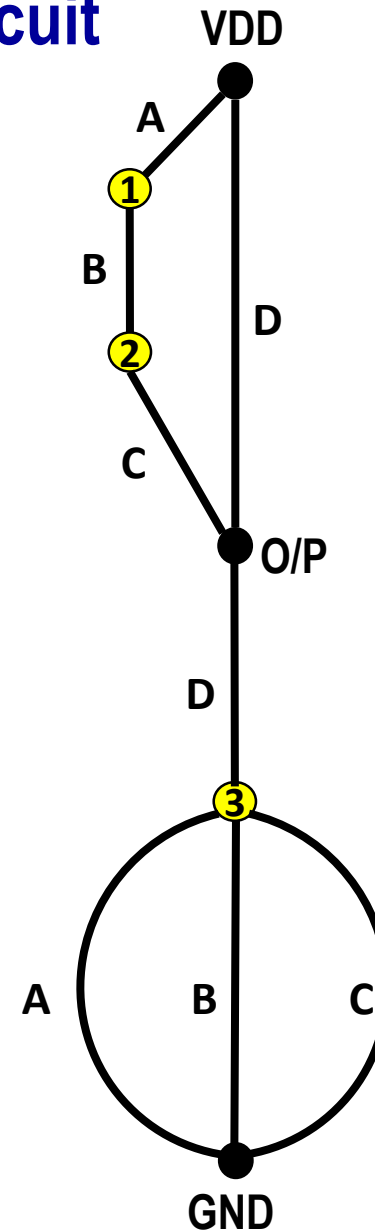
□ Draw the stick diagram of the following circuit
($f = \overline{D \cdot (A + B + C)}$)

□ Step4:- Identify the Euler path

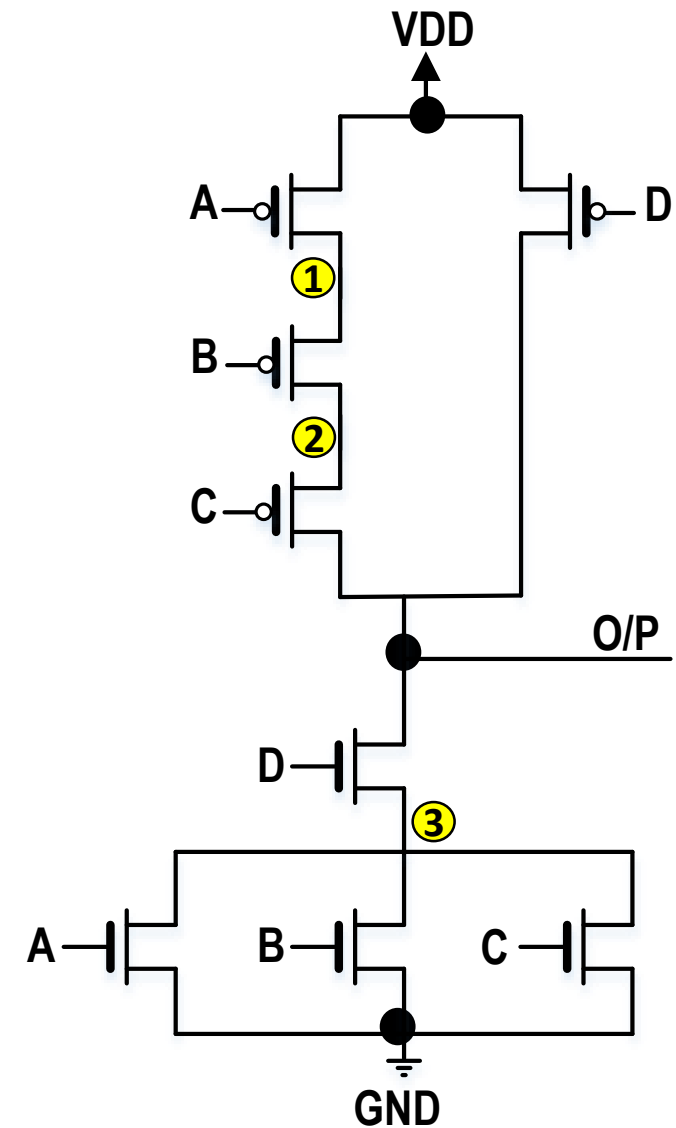
Condition of Euler Path

1. Nodes can be repeated but Paths cannot be repeated.

2. The path should be valid in both Pull-up and pull-down networks.



- A → B → C → D**



Exercise_04

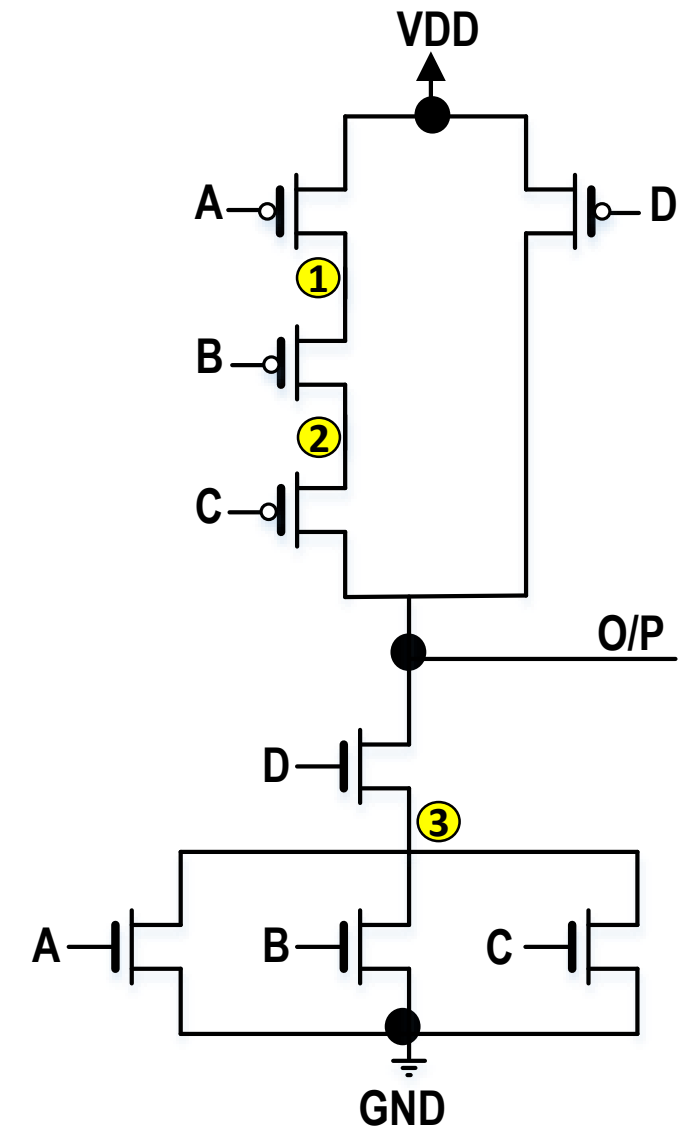
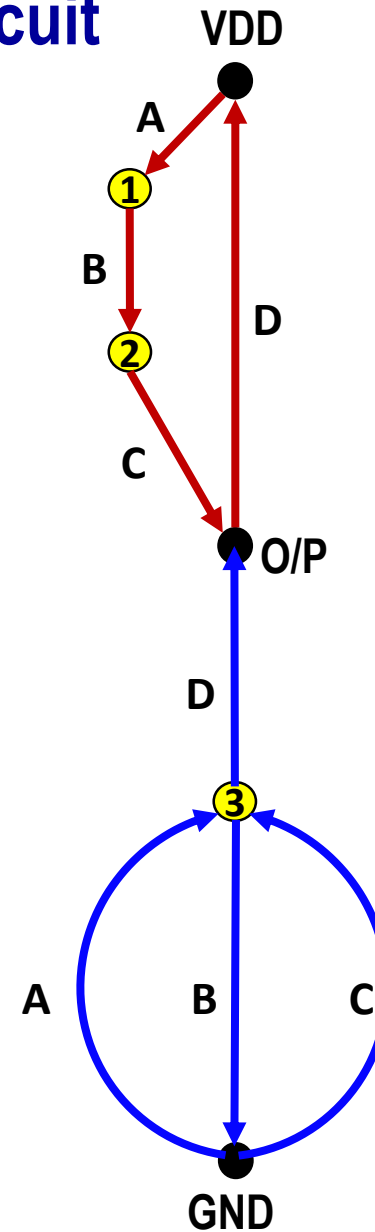


□ Draw the stick diagram of the following circuit
($f = \overline{D \cdot (A + B + C)}$)

□ Step4:- Identify the Euler path
Euler path for Pull-up network is

$A \rightarrow B \rightarrow C \rightarrow D$

Euler path for Pull-down network is
 $A \rightarrow B \rightarrow C \rightarrow D$



Exercise_04

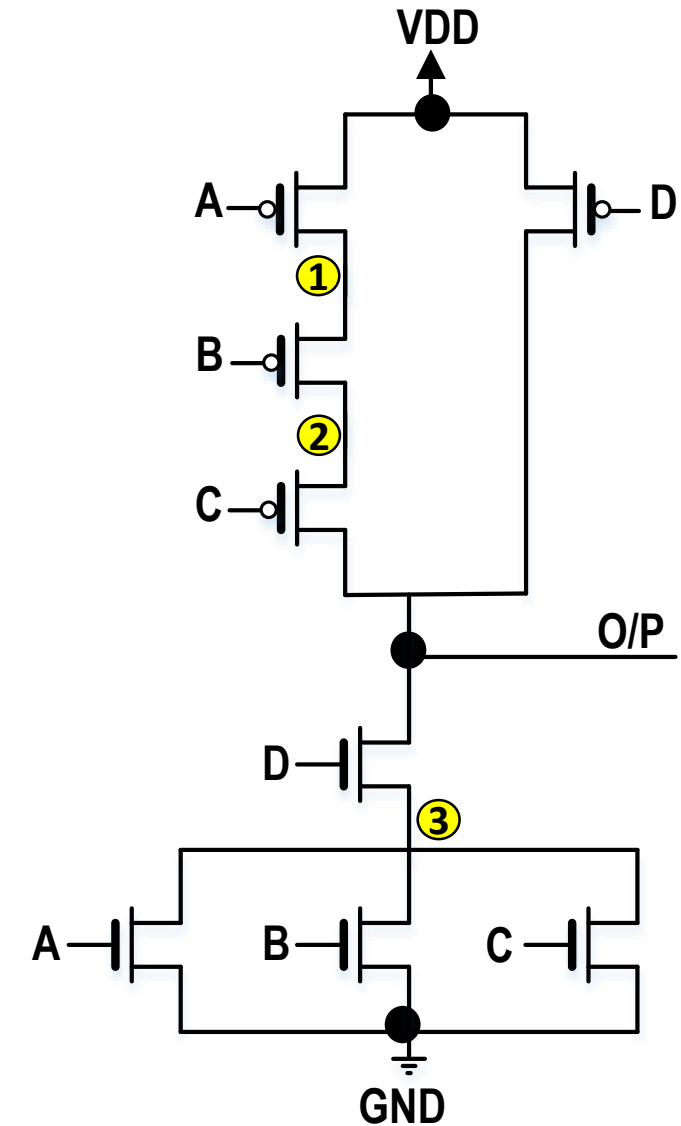
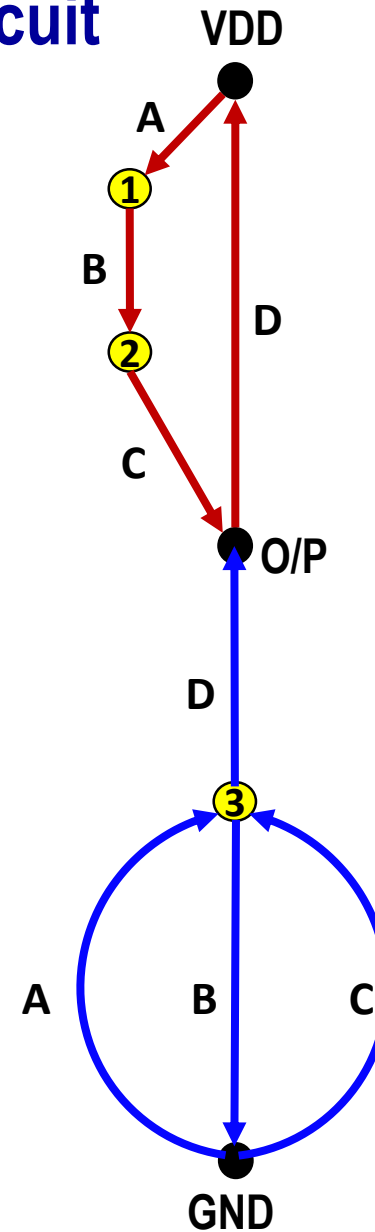


□ Draw the stick diagram of the following circuit

$$f = \overline{D \cdot (A + B + C)}$$

□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow C \rightarrow D$ (in order)

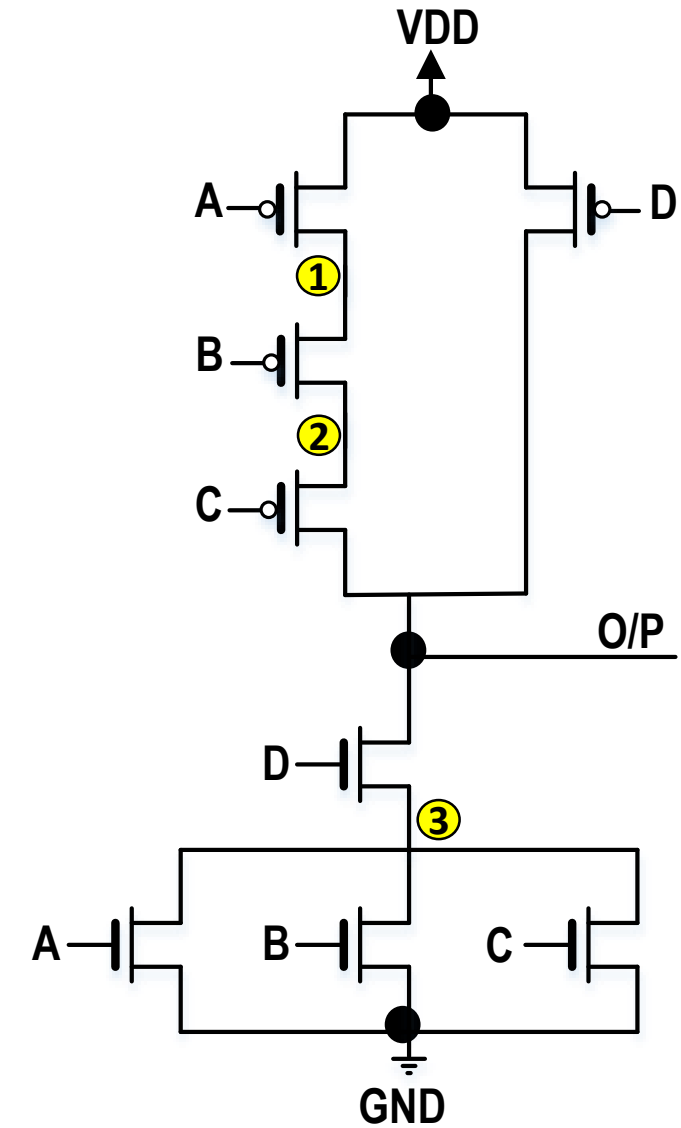
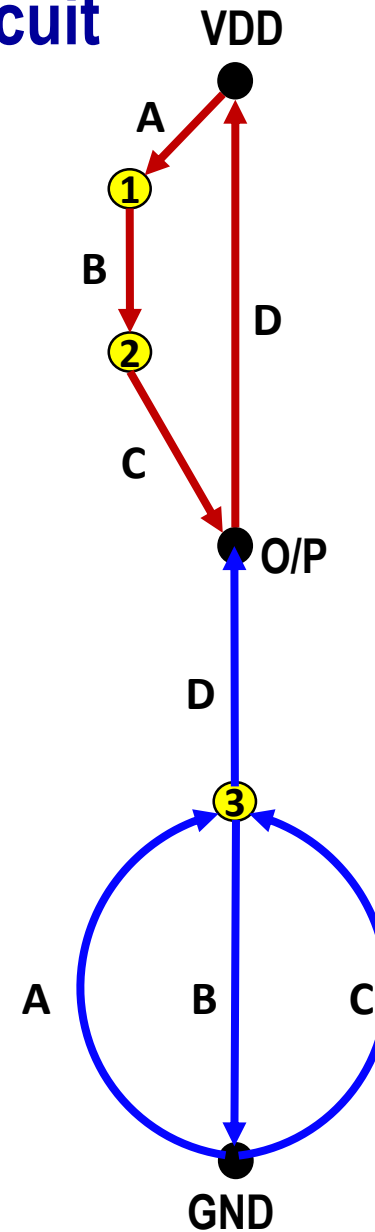
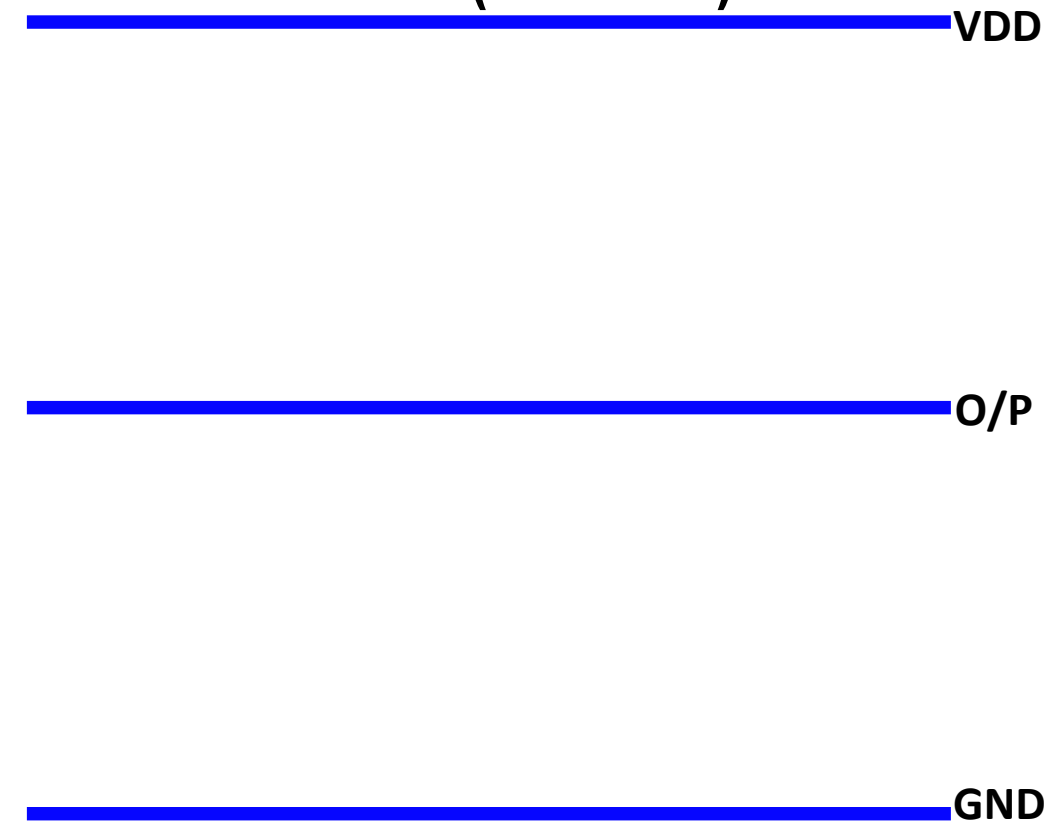


Exercise_04



□ Draw the stick diagram of the following circuit
($f = \overline{D \cdot (A + B + C)}$)

□ Step5:- Draw the stick diagram
 $A \rightarrow B \rightarrow C \rightarrow D$ (in order)

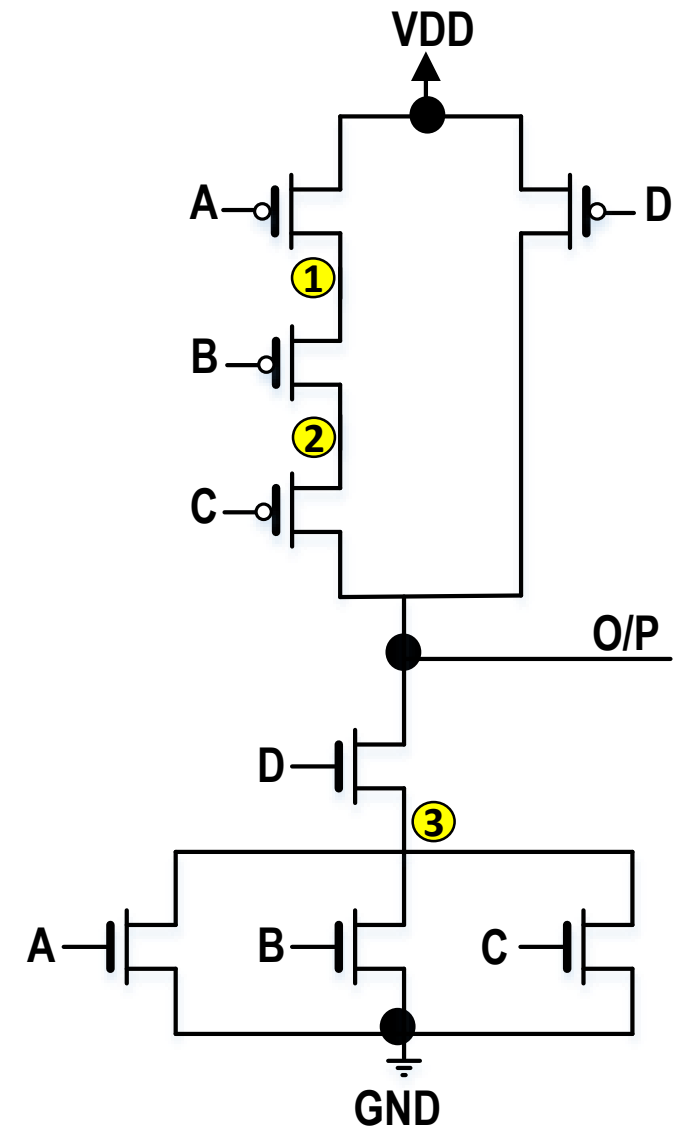
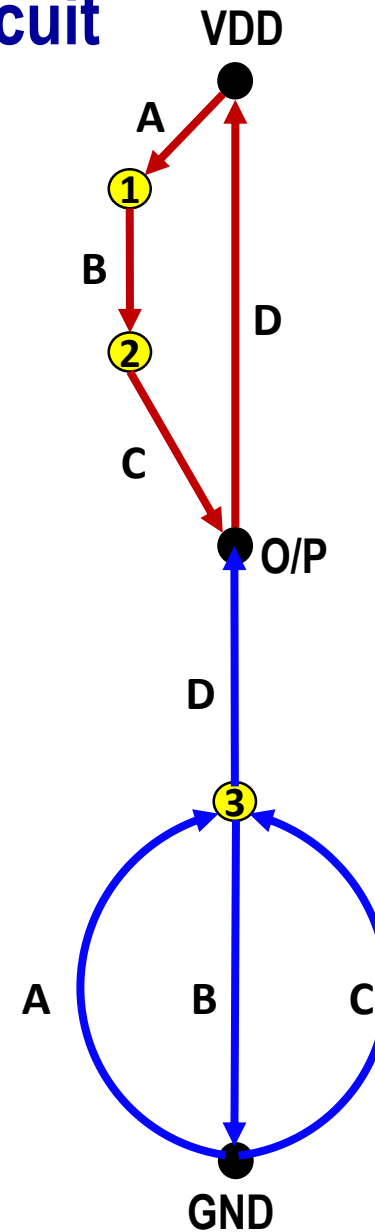
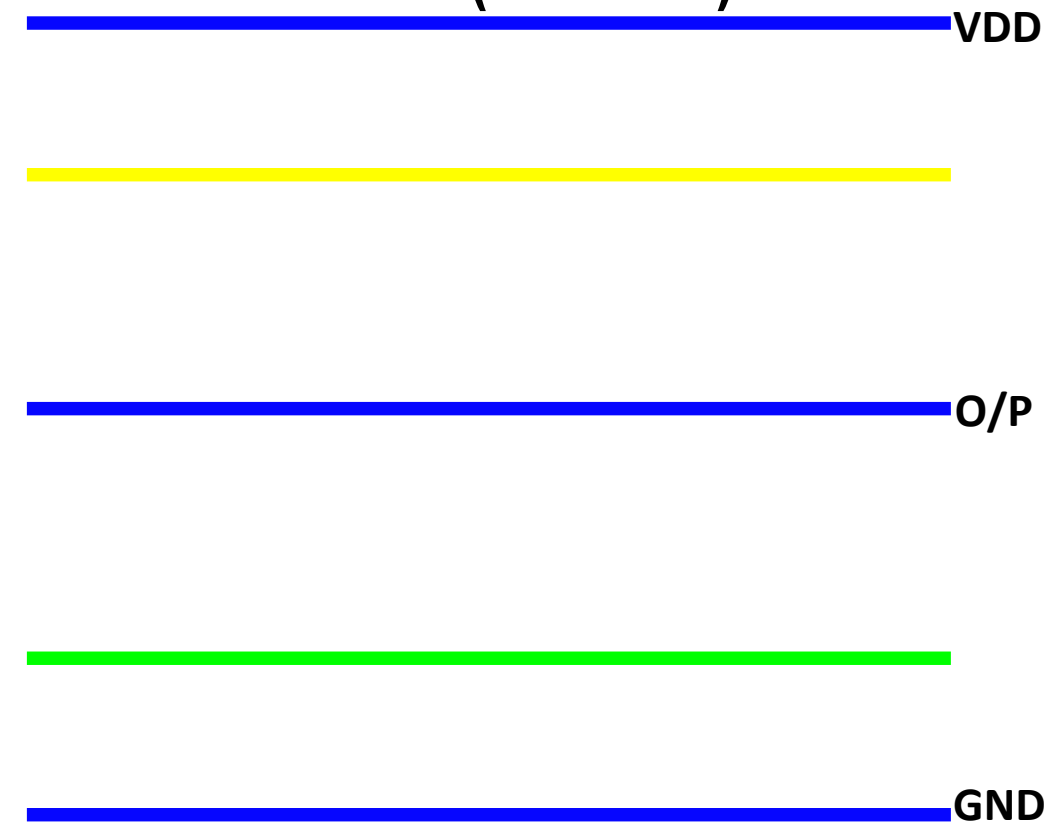


Exercise_04



□ Draw the stick diagram of the following circuit
($f = \overline{D \cdot (A + B + C)}$)

□ Step5:- Draw the stick diagram
 $A \rightarrow B \rightarrow C \rightarrow D$ (in order)



Exercise_04

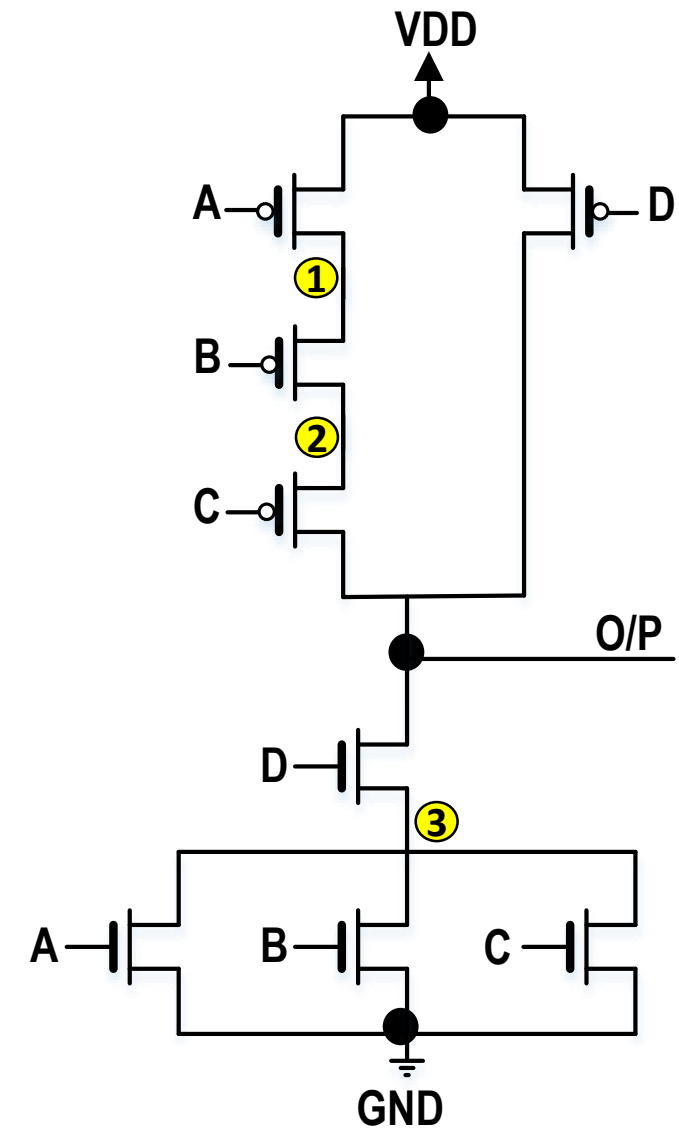
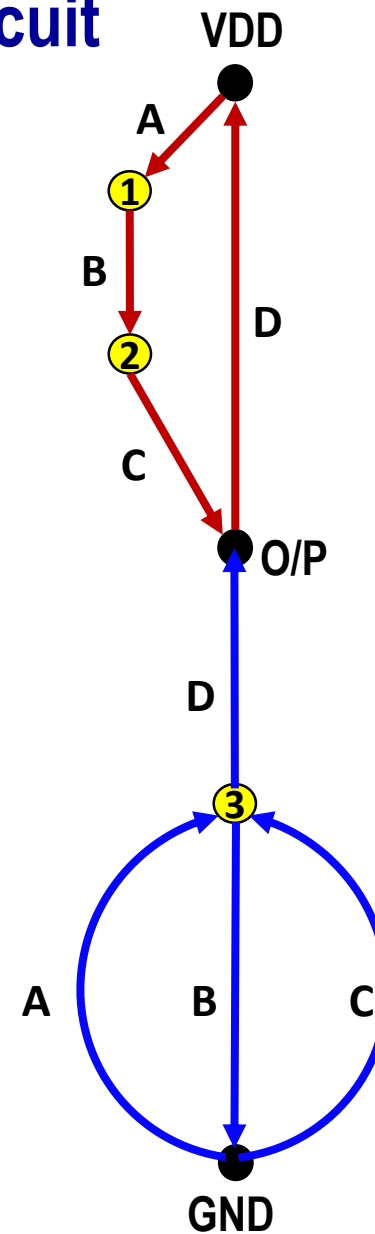
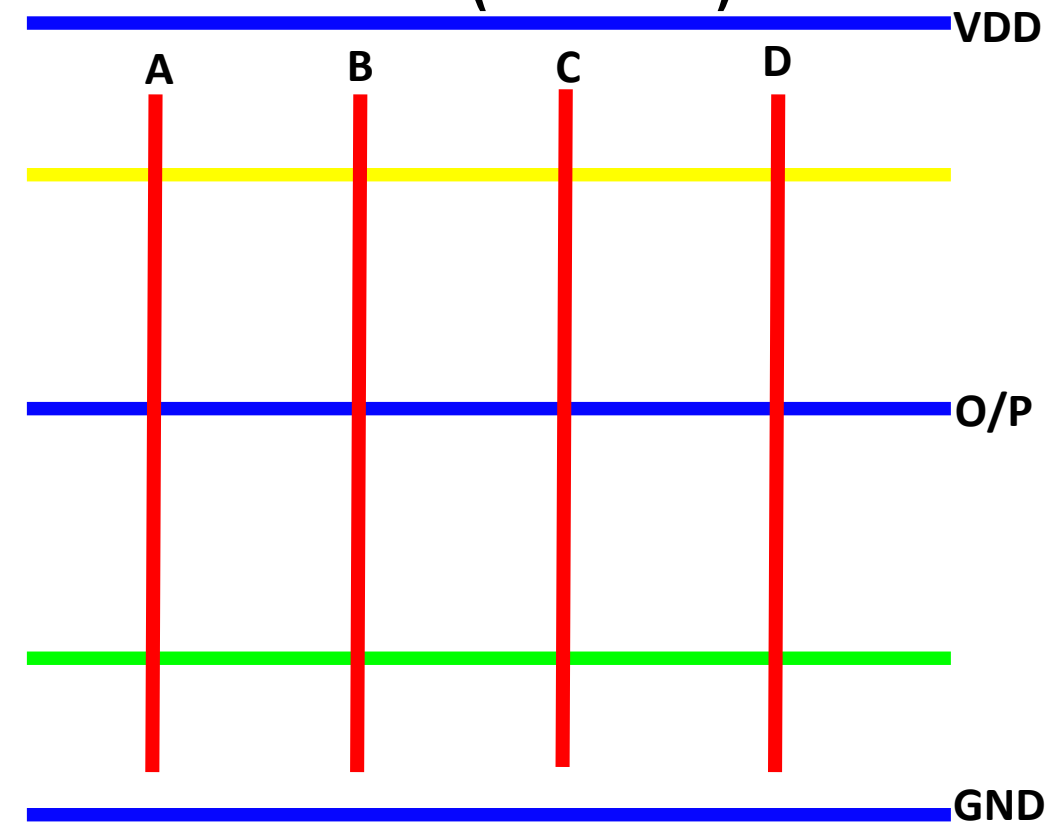


□ Draw the stick diagram of the following circuit

$$f = \overline{D \cdot (A + B + C)}$$

□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow C \rightarrow D$ (in order)



Exercise_04

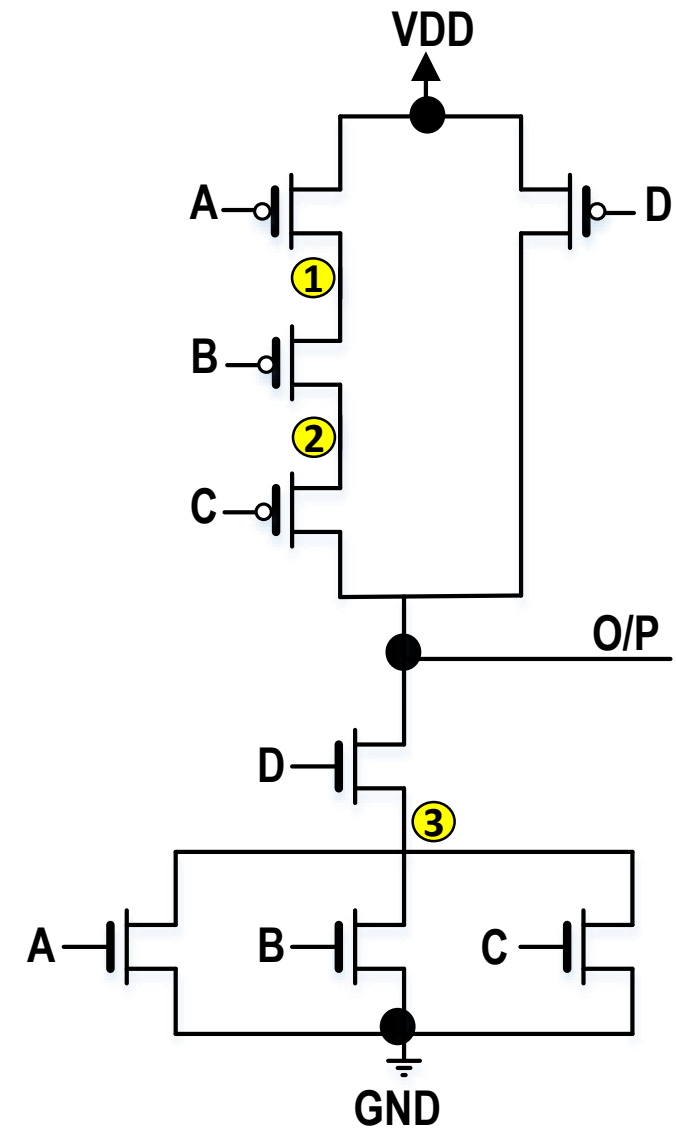
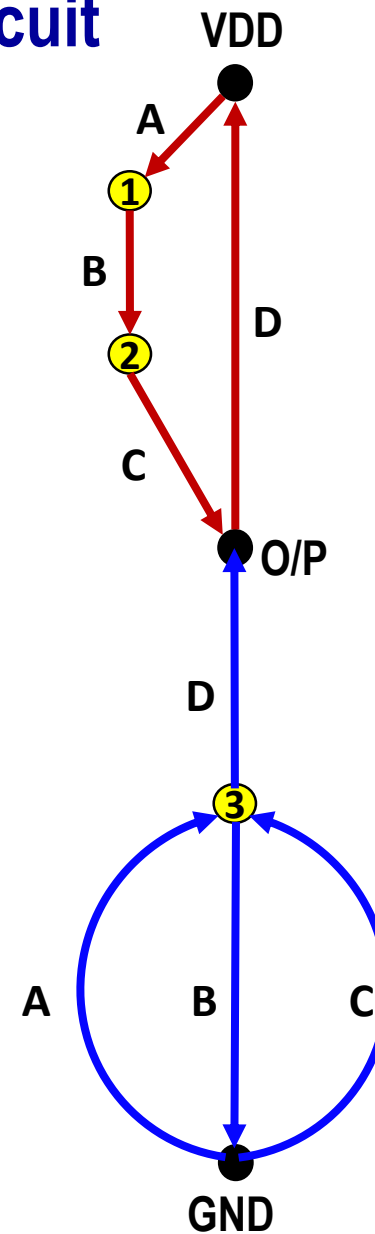
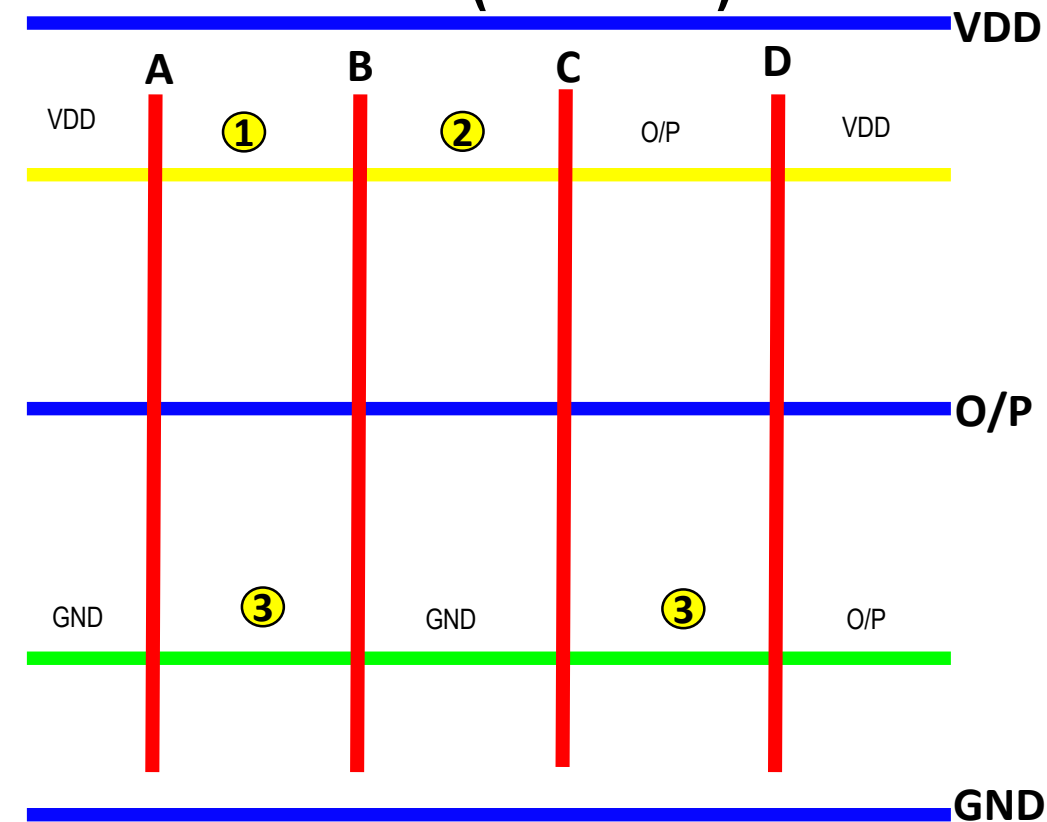


□ Draw the stick diagram of the following circuit

$$f = \overline{D \cdot (A + B + C)}$$

□ Step5:- Draw the stick diagram

$A \rightarrow B \rightarrow C \rightarrow D$ (in order)

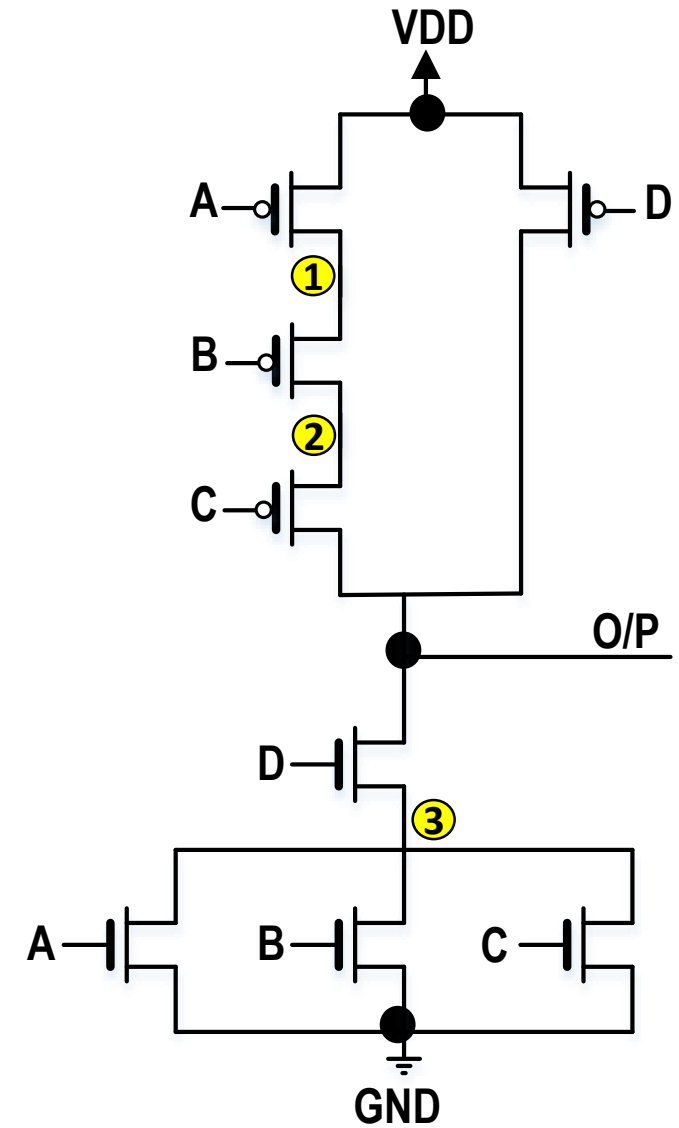
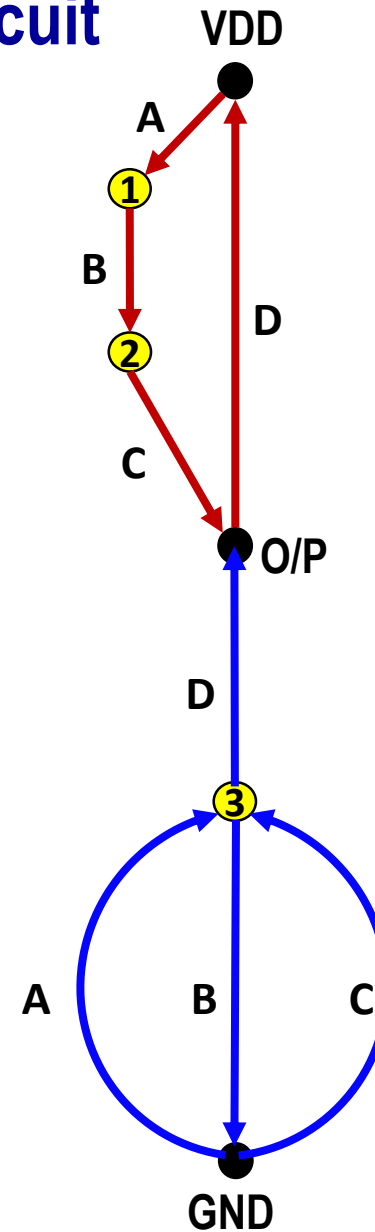
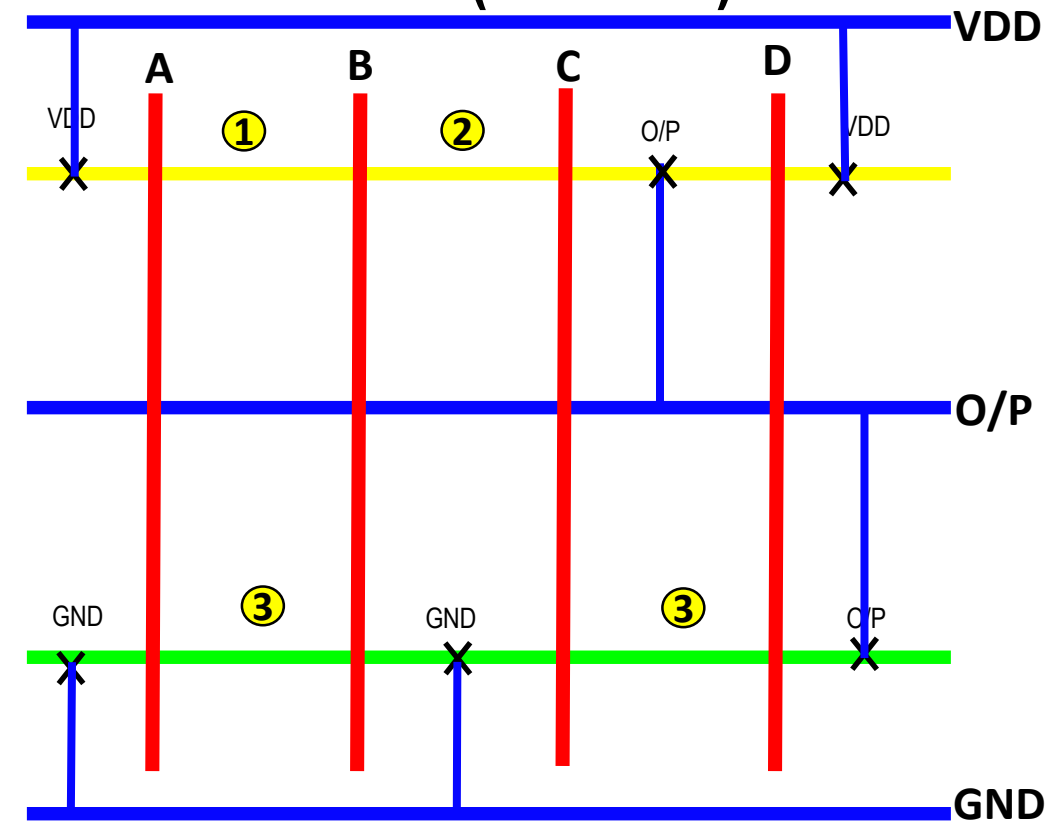


Exercise_04



□ Draw the stick diagram of the following circuit
 $f = \overline{D \cdot (A + B + C)}$

□ Step5:- Draw the stick diagram
 $A \rightarrow B \rightarrow C \rightarrow D$ (in order)

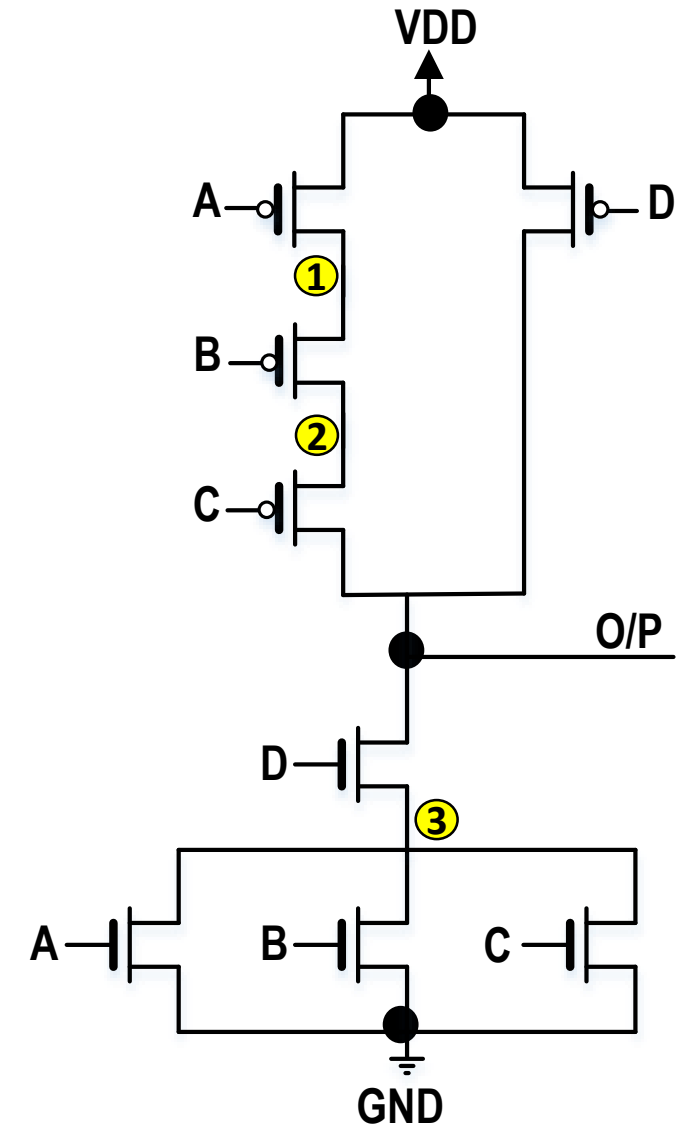
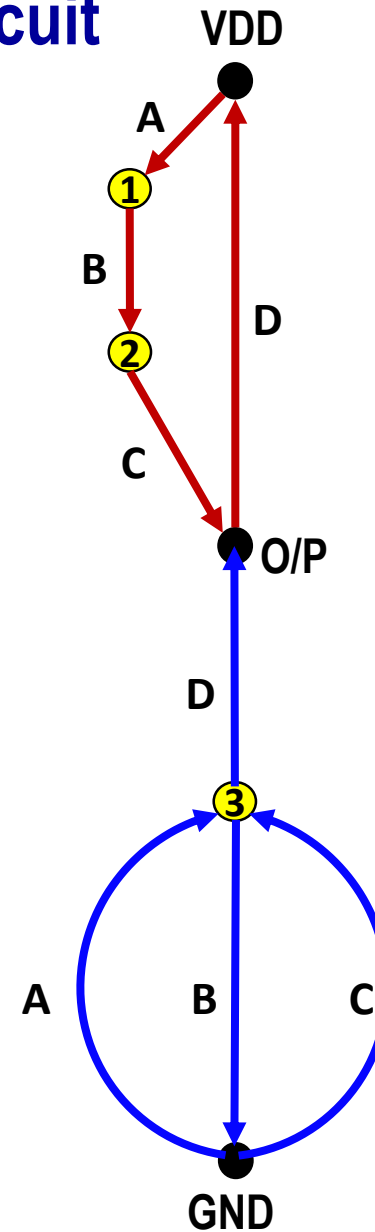
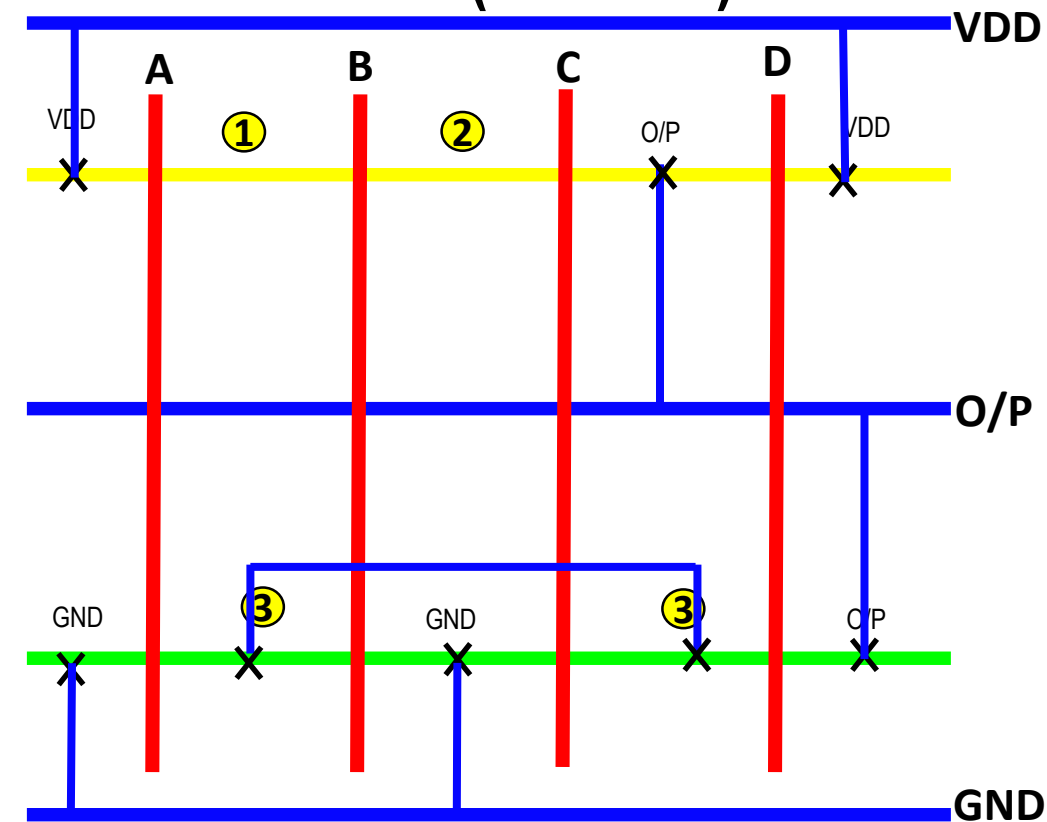


Exersice_04



□ Draw the stick diagram of the following circuit
 $f = \overline{(D \cdot (A + B + C))}$

□ Step5:- Draw the stick diagram
 $A \rightarrow B \rightarrow C \rightarrow D$ (in order)



Thanks!